
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

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ABSTRACT

Uniform manifold approximation and projection (UMAP) is among the most widely adopted neighbor embedding methods. The algorithm computes a low-dimensional embedding by optimizing a cross-entropy objective whose gradient decomposes into pairwise attractive and repulsive forces acting on high-dimensional data points. In this work, we analyze these forces to characterize their respective contributions to cluster formation and visualization, and we position UMAP relative to contemporaneous methods. The repulsive term accentuates dissimilarity, governing cluster boundaries and inter-cluster separation. The attractive term is more intricate: for sufficiently small pairwise distances, an attractive gradient update can induce local expansion rather than contraction, so a single force manifests as both attraction and repulsion in the low-dimensional map. This dichotomy accounts formally for the necessity of learning-rate annealing and motivates the asymmetric treatment of the attractive and repulsive terms. We further demonstrate that modifying the functional form of attraction increases the consistency of cluster formation under random initialization. Collectively, this analysis renders UMAP and related embedding methods more interpretable, more robust, and more accurate.

1 Introduction

Contemporary data analysis is increasingly dominated by high-dimensional observations whose intrinsic structure is not directly accessible. Dimensionality reduction (DR) techniques address this regime by constructing low-dimensional representations that preserve salient structure and admit direct visualization. The two most widely used methods, *t*-distributed stochastic neighbor embedding (Maaten and Hinton, 2008) and uniform manifold approximation and projection (UMAP) (McInnes et al., 2018), are both governed by attraction–repulsion dynamics: an optimization procedure that contracts the pairwise distances of similar points while expanding those of dissimilar points. As unsupervised procedures, they require no label supervision; instead, they recover and preserve the intrinsic structure of the data by encoding local (attractive) and global (repulsive) pairwise relationships as forces in the embedding. These properties make such methods applicable to clustering (Becht et al., 2019), exploratory data analysis (Fleischer and Islam, 2020), anomaly detection in semiconductor manufacturing (Fan et al., 2021), visual search (González-Márquez et al., 2024), time series analysis (Altin and Cakir, 2024), the study of representation convergence (Huh et al., 2024), and outlier image detection (Islam and Fleischer, 2024), all settings in which visualizing latent structure in unlabeled data is essential. By estimating embeddings in a data-driven, label-free manner, DR illustrates how unsupervised objectives can compress complex data into interpretable low-dimensional representations.

Extending the attraction–repulsion principle, a family of subsequent methods (Amid and Warmuth, 2019; Agrawal et al., 2021; Wang et al., 2021; Narayan et al., 2021; Yang et al., 2022; Wang et al., 2025; Kury et al., 2025) has been developed, each parameterized to emphasize particular structural properties of the data. Notwithstanding their broad applicability, these methods rely substantially on heuristic design choices whose effect on the resulting embedding is not always interpretable, and DR is furthermore subject to distortions that cannot be eliminated in principle (Chari and Pachter, 2023). A more rigorous account of the underlying algorithms is therefore required, so that practitioners can interpret embeddings correctly, distinguish genuine structure from artifacts, and tune performance in a principled manner. In practice, these algorithms attain compact clusters through a combination of mechanisms, including specific initialization schemes, learning-rate schedules, and kernel-function tuning. The dynamics of the attractive and repulsive forces that drive cluster formation, however, have not been systematically characterized. Moreover, the effective tunable

degrees of freedom remain implicit within the functional forms of the kernel and loss, which obscures the mechanistic behavior of the algorithms.

In this paper, we decompose each per-iteration force into a scalar coefficient acting along the pairwise displacement vector and extract the resulting functional forms, which we term the attraction and repulsion shapes. This decomposition shows that the necessity of learning-rate annealing, the difficulty of obtaining consistent output under random initialization, and the mechanism of cluster formation are all governed primarily by attraction, whereas the repulsive forces principally regulate inter-cluster distances. Our specific contributions are as follows:

1. We define the attraction and repulsion shapes as the scalar gradient coefficients of the attractive and repulsive forces, derive explicit conditions under which a single update contracts or expands the pairwise distance, and thereby provide a mechanistic reinterpretation of these algorithms (Section 4).
2. We prove that UMAP’s attraction shape induces both contraction and expansion of pairwise distance depending on the current separation, and, by comparing the attraction shapes of several algorithms, we explain how this property determines the requirement for a learning-rate annealing schedule (Section 5).
3. We modify the attraction shape and show empirically that the modification improves the consistency of the embedding under random initialization, providing evidence that UMAP encodes a unique underlying structure (Section 5.1).
4. By analyzing the repulsion shapes, we clarify the mechanisms of cluster formation and the regulation of inter-cluster distance (Section 5.2).
5. We compare the attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG- t -SNE, identify their formal correspondences and distinctions, and characterize the numerical stability of each algorithm (Section 6).

We develop the main text around UMAP, now a de facto standard across many fields, and introduce the other algorithms where they are pertinent. Detailed treatments of the individual algorithms are deferred to the appendix. The code used in this research will be released upon acceptance.

2 Related Works

Modern iterative graph-based neighbor embedding algorithms originate with stochastic neighbor embedding (SNE) (Hinton and Roweis, 2002) and its heavy-tailed extension based on the Student- t distribution (t -SNE) (Maaten and Hinton, 2008). Both methods operate on a dense affinity graph in which every point maintains a pairwise relation to all others, irrespective of similarity, and the graph weights are normalized so as to define a probability distribution. Concurrent methods, including locally linear embedding (Roweis and Saul, 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi, 2002), instead operate on a k -nearest-neighbor (k -NN) graph of pairwise interactions. These are spectral methods, and their solutions are obtained by eigenvalue decomposition of the associated graph operator. Tang et al. (2016) subsequently incorporated the k -NN graph into an iterative optimization procedure and dispensed with normalization in the low-dimensional representation. McInnes et al. (2018) extended this approach in UMAP by removing the normalization step entirely, in both the high- and low-dimensional representations, and computing the embedding from pairwise interactions alone. The corresponding optimization applies an explicit attractive force to preserve local neighborhoods and a repulsive force to separate dissimilar points. Building on these formulations, methods such as PaCMAP (Wang et al., 2021) and NEG- t -SNE (Damrich et al., 2023) have since been proposed. For a recent survey, see de Bodt et al. (2025).

A substantial body of work has sought to formally characterize the relationships among these algorithms. An early analysis of SNE established that data well separated into clusters in the original space remain well separated in the embedding space (Shaham and Steinerberger, 2017). An analogous analysis of t -SNE by Linderman and Steinerberger (2019) proved that the number of clusters recovered in the embedding space lower-bounds the number of clusters in the original space, a result refined by subsequent characterizations (Arora et al., 2018; Cai and Ma, 2022; Linderman and Steinerberger, 2022). Because t -SNE and UMAP derive from a common underlying framework yet yield markedly different visualizations, establishing the precise correspondence between them has been a central line of inquiry (Böhm et al., 2022; Damrich et al., 2023; Draganov et al., 2023). Böhm et al. (2022) conjectured that Laplacian Eigenmaps, ForceAtlas2 (Jacomy et al., 2014), UMAP, and t -SNE constitute instances along a single continuous spectrum of methods. Consistent with this view, LE and t -SNE are linked through the early-exaggeration phase (Cai and Ma, 2022), and UMAP and t -SNE are related through contrastive estimation (Damrich et al., 2023). Concurrently, Hu et al. (2023) independently established the connection between contrastive learning and SNE. More recent analyses account for dimensionality reduction from a probabilistic perspective (Ravuri et al., 2023; Ravuri and Lawrence, 2024), through kernel-based techniques (Draganov and Dohn, 2023), and via information geometry (Kolpakov and Rocke, 2024).

3 Uniform Manifold Approximation and Projection

Given a dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$, UMAP constructs a high-dimensional affinity graph from a pairwise relation $p_{i,j} = f_h(d(x_i, x_j))$, where f_h maps the input distance to an affinity, typically in $[0, 1]$ (see Appendix H for the precise construction).

The affinities of the low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ are given by the differentiable function

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

where the parameters a and b control the density profile of the mapping and are determined by a nonlinear least-squares fit of q_{ij} to the piecewise target function

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d, \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

where the parameter m_d sets the minimum admissible separation between the two nearest low-dimensional points. UMAP minimizes the cross-entropy between the high- and low-dimensional affinities,

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

The first term contributes an attractive force and the second term a repulsive force. Rather than evaluating the full loss over all pairs at each iteration, UMAP employs negative sampling (Mikolov et al. 2013, Tang et al. 2016): for each edge with $p_{ij} > 0$, designated a positive edge, a fixed number of edges are drawn at random and designated negative edges. The attractive force acts on the positive edge via the gradient-ascent updates

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

and the repulsive force acts on each negative edge via the update

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ denotes the learning rate and t indexes the optimization step. For negative edges, the endpoint y_j is held fixed and only y_i is updated. A detailed analysis of UMAP’s loss is given in Damrich and Hamprecht (2021).

4 Attraction and Repulsion Shapes

The updates (4, 6) admit a compact reformulation: for a Euclidean metric, each gradient $\nabla_{y_i^t} \log(q_{ij})$ (respectively $\nabla_{y_i^t} \log(1 - q_{ij})$) factors into a scalar coefficient that depends only on the pairwise distance $\|y_i - y_j\|_2$, multiplied by the displacement vector $(y_i - y_j)$. We refer to this scalar coefficient as the attraction (respectively repulsion) shape. Although we instantiate the construction on UMAP, the formalism applies to any method whose optimization is driven by pairwise attractive and repulsive forces.

Writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$ with $\zeta = \|y_i - y_j\|_2$, the positive-edge update becomes

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ is the attraction shape, and we have used the antisymmetry $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$ that holds under the Euclidean metric. Analogously, the negative-edge update is reformulated as

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape.

Decompositions of this form have appeared previously, e.g., in (McInnes et al. 2018, Agrawal et al. 2021, Draganov and Dohn 2023), though their formulation and use differ from ours. The original UMAP paper (McInnes et al. 2018)

employed it as a computational device for efficient gradient evaluation, and Draganov and Dohn (2023) used it to compare algorithms from a kernel perspective; in both cases the objective was the derivative itself. Here, by contrast, we treat the scalar coefficients as functions in their own right, whose functional form may be specified independently of any particular loss. A related decomposition was introduced earlier by Agrawal et al. (2021), expressed as a scalar coefficient multiplied by the unit displacement vector $(y_i - y_j)/\|y_i - y_j\|_2$ so as to isolate the magnitude of the force. As the following analysis shows, our shape-based decomposition is more informative than the force magnitude alone.

4.1 Conditions for Attraction and Repulsion

In this section we derive, directly from the update equations (7)-(10), the precise conditions under which a single update contracts or expands a pairwise distance. The following proposition characterizes contraction of the distance between the endpoints y_i^t and y_j^t of a positive edge:

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

Here the product λf_a acts as the effective attraction shape; in most cases, however, f_a alone suffices to draw meaningful conclusions about the embeddings. For a negative edge, the following proposition characterizes expansion of the distance between the endpoints y_i^t and y_j^t :

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

We note that including a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ in Eq. (10) leaves this conclusion unchanged. These conditions constitute a per-iteration certificate of guaranteed contraction or expansion; they should not be conflated with the learning-rate schedule, since the shapes can be designed so that contraction and expansion hold independently of any learning-rate decay.

Figure 1(a) illustrates the effect of representative values of $f_a < 0$ and $f_r > 0$ on a pair of points. The repulsive case is immediate: any positive coefficient increases the separation. The attractive case is more subtle, because f_a encodes both attractive and repulsive behavior. For $f_a \in (-1.0, 0)$ the distance strictly decreases, with the maximal contraction (the pair becoming coincident) attained at $f_a = -0.5$; beyond this value the endpoints cross over. For any $f_a < -1.0$ the distance instead increases. Although each update acts on a single pair, the aggregate of these local updates determines the global structure of the embedding.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Using the gradient decomposition and the distance form (1), the attraction and repulsion shapes are given by

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. This section analyzes the default shapes of UMAP, which are controlled primarily by the parameters a and b , and examines the effect of perturbing them. Unless stated otherwise, the analysis assumes $b \leq 1$, the regime in which the attraction shape is strictly increasing and hence invertible (see Appendix A for the derivation).

Figure 1(b) shows the default attraction shape of UMAP ($a = 1.58, b = 0.89$), which diverges to $-\infty$ in magnitude as $\zeta \rightarrow 0$. By Proposition 4.1 the transition from contraction to expansion occurs precisely where λf_a^U crosses -1 , which happens as $\zeta \rightarrow 0$. Because the attraction shape is invertible, the distance at this transition, $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$, is well defined and identifies the minimum distance for contraction under attractive updates. Consequently, $\zeta > \zeta_{-1}$ yields contraction while $0 < \zeta < \zeta_{-1}$ yields expansion, in direct contradiction to the intuition that attractive updates uniformly reduce pairwise distances.

When ζ_{-1} is large, neighboring points oscillate between contraction and expansion across iterations, producing diffuse cluster boundaries. The objective of obtaining sharp boundaries can therefore be restated as driving $\zeta_{-1} \rightarrow 0$. Since ζ_{-1} decreases monotonically with λ , this requires annealing the learning rate to zero (Figs. 1(c,d)). The repulsion shape (Eq. 12), by contrast, is strictly positive and thus satisfies Proposition 4.2 for all ζ ; $f_r^U \rightarrow 0$ as $\zeta \rightarrow \infty$ and $f_r^U \rightarrow \infty$ as $\zeta \rightarrow 0$ (Fig. 1(e)).

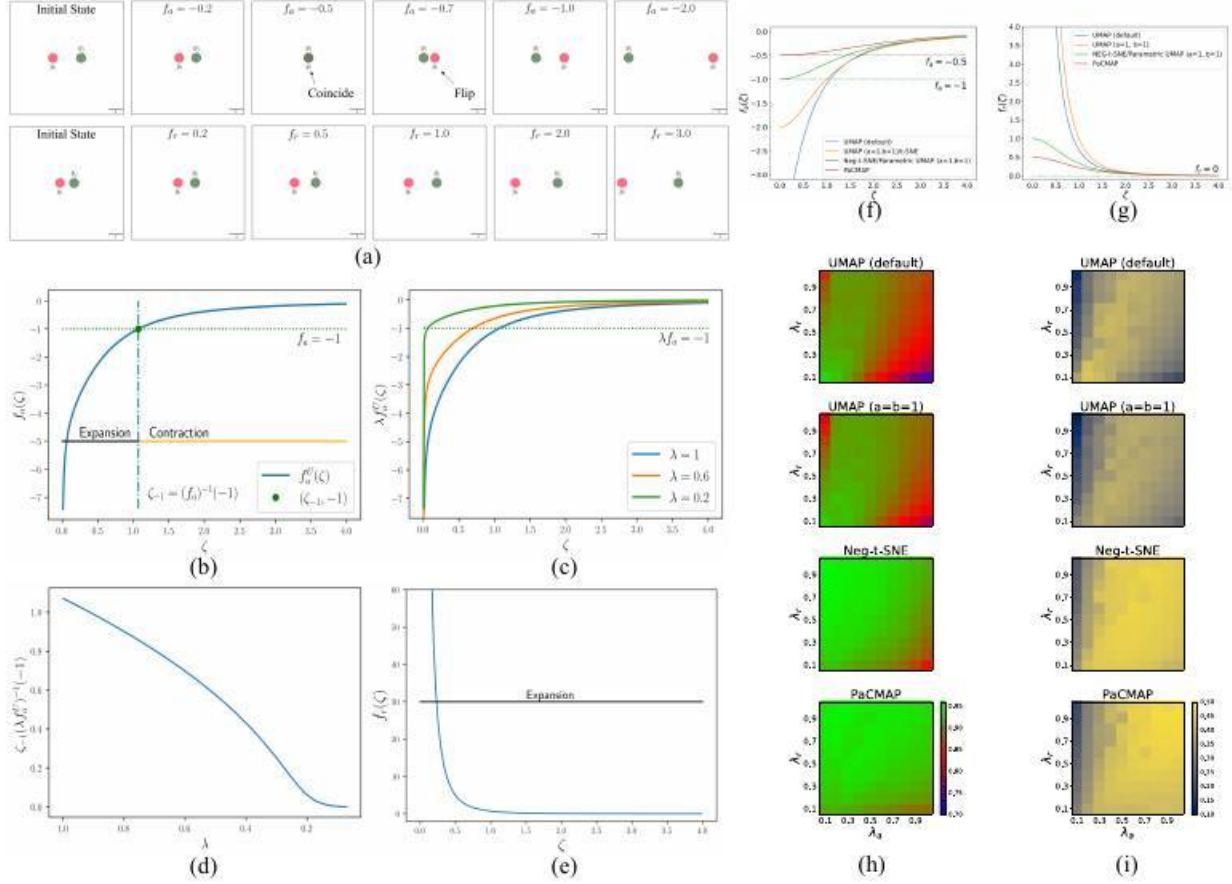


Figure 1: Attraction and repulsion shapes in UMAP. (a) Effect of distinct values of f_a (top) and f_r (bottom) on the separation of a single pair. (b) Attraction shape of UMAP as a function of the pairwise distance. (c) Effective attraction shape (λf_a) for a range of learning rates λ . (d) Minimum distance for contraction (ζ_{-1}) as a function of decreasing λ . (e) Repulsion shape of UMAP. (f) Attraction and (g) repulsion shapes for several embedding methods. (h) Trustworthiness and (i) Silhouette Score for each method as the constant learning rate is varied independently for attraction (λ_a) and repulsion (λ_r). Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

Comparing attraction shapes across algorithms reveals that only UMAP and t -SNE admit $f_a < -1$ (and correspondingly $\zeta_{-1} > 0$ at $\lambda = 1$; Fig. 1(f)). t -SNE resolves this by weighting each update by the corresponding affinity p_{ij} , whereas UMAP relies on learning-rate annealing. Neg- t -SNE and PaCMAP, in contrast, confine f_a to $[-1, 0]$ by construction, so they satisfy Proposition 4.1 for all $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$. PaCMAP moreover keeps its weighted f_a below 0.5 in magnitude at all distances, which precludes any endpoint crossover during attractive updates. The repulsion shapes exhibit an analogous distinction: only UMAP attains large values at small distances (Fig. 1(g)), where the shape is formally unbounded but is clipped in practice during optimization. Additional details on these and related methods are given in Appendix G.

We quantify the effect of these attraction and repulsion choices on the MNIST embedding by assigning separate constant learning rates to attraction (λ_a) and repulsion (λ_r) and sweeping each independently (Figs. 1(h,i)). For UMAP, a small λ_a (< 0.5 , so that $\zeta_{-1} \simeq 0$) yields superior structure preservation, as measured by trustworthiness (T) (Venna and Kaski 2001) in Fig. 1(h), and superior cluster separation, as measured by the silhouette score (SIL) (Rousseeuw 1987) in Fig. 1(i), while the entire range $\lambda_r \in [0, 1]$ is admissible. For Neg- t -SNE and PaCMAP, for which $\zeta_{-1} = 0$ by construction, the full parameter range is effective. These results confirm that UMAP requires $\zeta_{-1} \rightarrow 0$ for satisfactory embedding and clustering, a condition it enforces in practice through learning-rate annealing, whereas $\zeta_{-1} = 0$ holds by default in the newer algorithms. Further discussion, results, and illustrations on additional datasets are provided in Section 6 and Appendices B, G and I.

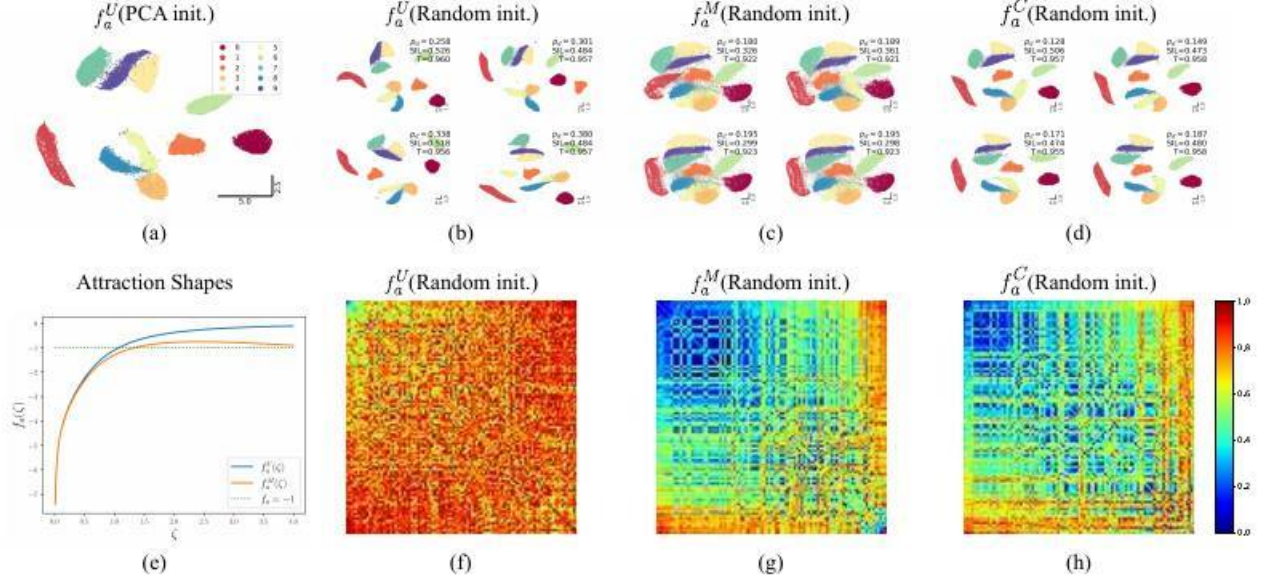


Figure 2: Effect of random initialization on distinct attraction shapes for the MNIST dataset. (a) Reference mapping obtained with PCA initialization. (b-d) The four mappings attaining the lowest Procrustes distance (p_d) to the reference in (a), for the (b) UMAP, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices aggregated over 100 runs for the (f) UMAP (0.78 ± 0.13), (g) modified (0.49 ± 0.21), and (h) composite (0.50 ± 0.20) attraction shapes. The diagonal entries (i, i) of each Procrustes matrix are sorted by their distance p_d to the reference (a), while the off-diagonal entries (i, j) report p_d between the i th and j th mappings. The matrices and the reported mean $p_d \pm \text{std}$ indicate that UMAP’s embeddings are not mutually self-similar, whereas the modified and composite attraction shapes induce a scale-invariant structure.

5.1 Improving UMAP’s Consistency under Random Initialization

The consistency of UMAP embeddings is contingent on the initialization (Kobak and Linderman 2021; Wang et al. 2021). The embedding is typically initialized from either the principal component analysis (PCA) projection of the data or the spectral decomposition of the high-dimensional graph, which yields consistent mappings despite the remaining sources of stochasticity. Under random initialization, by contrast, clusters frequently fail to form, or form in an arbitrary orientation on each run. The underlying cause is that when two high-dimensional nearest neighbors are initialized far apart, the attractive force is too weak to draw them together. This phenomenon, termed near-sightedness (Wang et al. 2021), is a direct consequence of the attraction shape: $|f_a^U|$ decays to zero with increasing distance, so that the net displacement vanishes in the limit, $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)|\zeta = 0$.

This suggests that “far-sightedness” can be induced by augmenting the attraction at large distances, so that distant neighbors are drawn together. To test this hypothesis, we augment UMAP’s attraction shape with an additive term that grows with distance:

$$f_a^M = f_a^U - \beta\zeta, \quad (13)$$

where $\beta \geq 0$ controls the strength of the added term (we set $\beta = 0.2$; Fig. 2(e)). Adding this term to the attraction shape is equivalent to introducing a regularizer in the attractive term of the loss, $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^3$. Beyond attracting distant pairs, the added term promotes the intermixing of points that facilitates convergence under random initialization, analogous to the early-exaggeration phase of t -SNE. We adopt the linear correction in (13) as the simplest such choice; alternative monotone corrections, such as $\log \zeta$ or ζ^p with $p \in \mathbb{R}_{\geq 0}$, are equally admissible.

We also consider a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases}. \quad (14)$$

The composite shape is designed to remove any distortion introduced by f_a^M by reverting to the original UMAP shape after the initial epochs. We next examine the effect of the modified and composite attraction shapes on DR under random initialization.

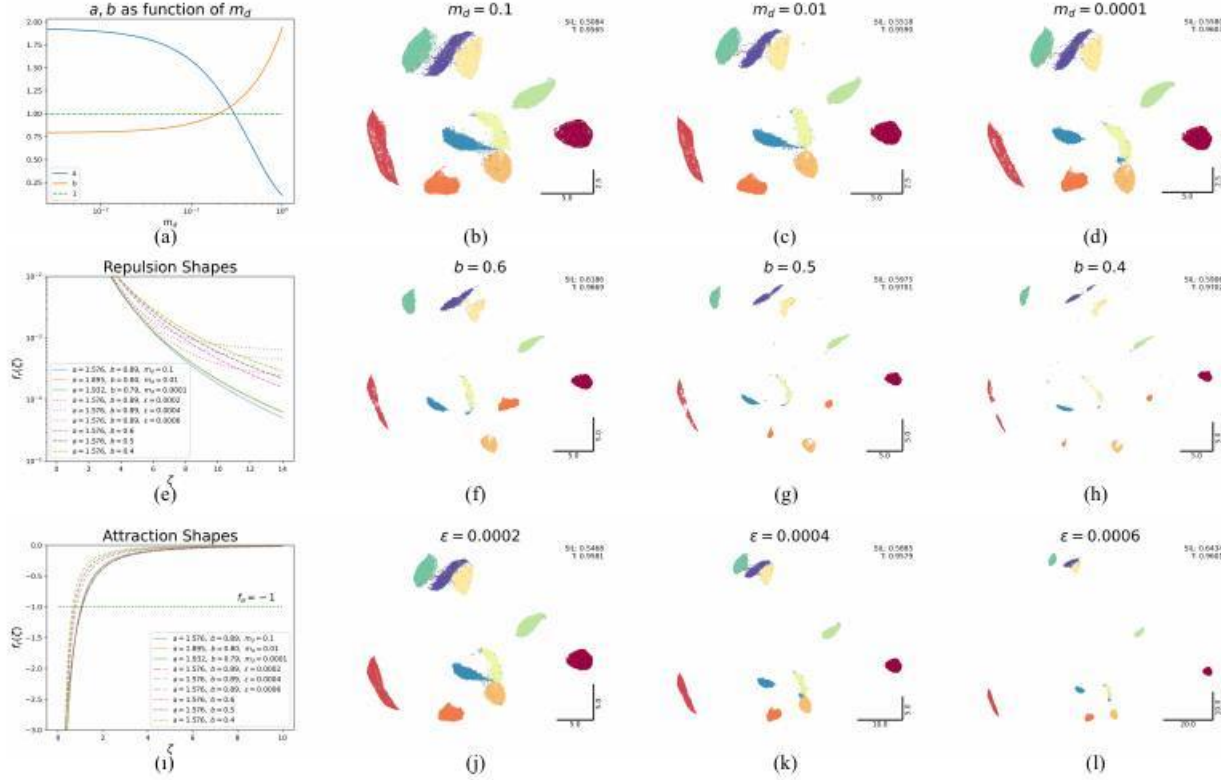


Figure 3: Control of inter-cluster distances on the MNIST dataset. (a) Determining a and b by varying the low-dimensional distance m_d constrains the accessible region of parameter space. (b-d) UMAP embeddings obtained by setting m_d to 0.1, 0.01, and 0.0001, respectively, exhibit negligible gains in cluster compactness. (e) Repulsion shapes for the corresponding parameter settings. (f-h) Amplifying repulsion by directly decreasing b produces more compact clusters and resolves clusters that were otherwise absent. (i) Attraction shapes for the corresponding parameter settings. (j-l) Amplifying repulsion by adding a small positive constant (ϵ) to the repulsion shape increases the inter-cluster distance.

We first computed a PCA-initialized embedding of the MNIST dataset (Fig. 2(a)) to serve as a reference. For each attraction shape, we then generated 100 embeddings under independent Gaussian random initializations. To quantify agreement, we use Procrustes analysis (Gower 1975), which optimally aligns two point clouds under scaling, translation, rotation, and reflection (see Appendix D.1 for the definition). Each randomly initialized embedding is aligned to the PCA-initialized reference, and the residual after alignment is quantified by the Procrustes distance (p_d). Figure 2(b) shows the four embeddings with the lowest p_d : the cluster shapes are consistent, but their spatial placements are not. The modified and composite attraction shapes (Figs. 2(c) and (d), respectively) yield markedly more consistent cluster placements.

To characterize the placements more finely, we form the Procrustes matrix, whose diagonal entries record p_d to the PCA-initialized reference (sorted in ascending order) and whose off-diagonal entries record p_d between pairs of randomly initialized mappings (see Appendix D.1); this construction is analogous to a self-similarity matrix (Foote 1999). The embeddings produced by the default UMAP attraction shape are mutually dissimilar (Fig. 2(f)), whereas those produced by the modified (Fig. 2(g)) and composite (Fig. 2(h)) shapes are strongly mutually similar. This indicates that UMAP encodes a unique underlying structure that is independent of the initialization, but that recovering this structure in the low dimension may be impeded by the vanishing attraction at large distances.

5.2 Cluster Formation and Compactness

The parameter that ostensibly controls cluster formation in UMAP is the minimum distance m_d , which enters through Eq. 2. Varying m_d , however, traces out only a restricted locus in the (a, b) plane (Fig. 3(a)), so reducing m_d typically yields embeddings with no appreciable improvement (Figs. 3(b-d)). The reason is that m_d exerts little influence on the repulsion shape (Fig. 3(e)). An alternative is to vary a and b directly. Decreasing a increases repulsion but attenuates

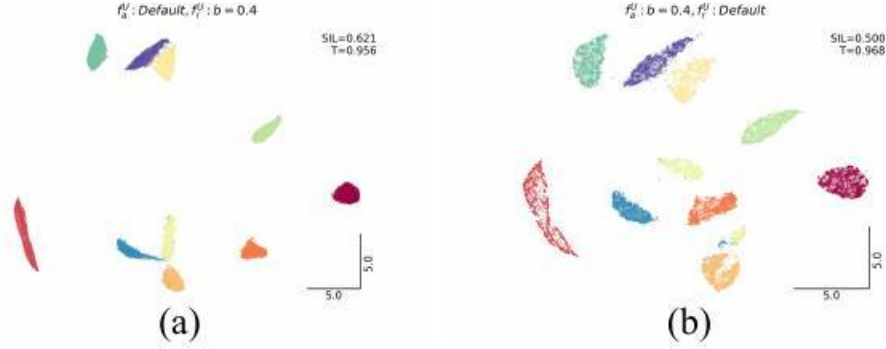


Figure 4: Embedding of the MNIST dataset under selective modification of a single shape: (a) the default attraction shape combined with a repulsion shape evaluated at $b = 0.4$, and (b) the default repulsion shape combined with an attraction shape evaluated at $b = 0.4$. Configuration (a) reproduces the clusters of default UMAP with increased compactness, whereas configuration (b) develops additional intra-cluster structure and resolves new clusters.

attraction more rapidly, exacerbating near-sightedness. Decreasing b , by contrast, affords finer control (Figs. 3(e,i)). Figures 3(f-h) show that setting b to 0.6, 0.5, and 0.4 progressively increases the inter-cluster distance and fragments existing clusters. This fragmentation arises because the kernel becomes increasingly heavy-tailed as b decreases; heavy-tailed kernels yield smaller, more distinct clusters (Van der Maaten and Hinton 2008, Kobak et al. 2019, and Kobak et al. (2019) details how b modulates the tail in UMAP. While this procedure separates all ten MNIST labels into distinct clusters, it does not isolate the contributions of attraction and repulsion, because decreasing b amplifies repulsion more strongly than decreasing m_d but simultaneously reshapes the attraction profile.

To isolate the effect of repulsion, we hold the attraction shape fixed and perturb the repulsion shape alone by an additive positive constant (ε):

$$f_r^M = f_r^U + \varepsilon, \quad (15)$$

holding a and b fixed. This perturbation is equivalent to adding a regularizer to the repulsive term of the loss, $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \varepsilon/2 \|y_i - y_j\|_2^2$, and attains stronger repulsion than is otherwise accessible (Fig. 3(e)). Figures 3(j-l) show that the inter-cluster distance increases monotonically with ε . The resulting clustering, however, is qualitatively similar to that obtained by varying m_d , yielding only a loose separation of the labels as ε grows. Crucially, ε leaves the attraction shape unchanged, and varying m_d traces essentially the same family of attraction shapes (Fig. 3(i)); together these observations indicate that cluster formation is governed predominantly by attraction.

To disentangle the mechanism of cluster formation, we modify either the attraction shape or the repulsion shape in isolation, evaluating one at $b = 0.4$ while holding the other at its default (Fig. 4). With the default attraction and increased repulsion ($b = 0.4$), no new structure emerges, though the clusters become more compact than in the original UMAP (Fig. 4(a)). With increased attraction ($b = 0.4$) and the default repulsion, by contrast, the embedding develops additional intra-cluster structure, and several existing clusters split into smaller ones (Fig. 4(b)). This dissociation establishes that attraction drives cluster formation, whereas repulsion controls compactness, either by forming smaller clusters or by increasing the inter-cluster distance, depending on the repulsion shape. In Appendix G.3 we show that LocalMAP embeddings exploit the same attraction–repulsion interplay to separate clusters.

6 Comparison to NEG- t -SNE

The choice $a = b = 1$ is common across dimensionality reduction algorithms, as it renders the gradients bounded and thereby stabilizes the optimization. Damrich et al. (2023) examined this setting for UMAP and proposed Neg- t -SNE, which combines a stable gradient with more compact clustering even under a constant learning rate of 1. Parametric UMAP (Sainburg et al. 2021), in turn, was originally formulated with the UMAP loss but subsequently adopted a numerically stable modified cross-entropy loss (Shi et al. 2023) with a log-sigmoid kernel. Analyzing both formulations shows that they converge to the same objective from distinct starting points, which yields the following equivalence:

Proposition 6.1. *Neg- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

github.com, merged on April 26, 2022

With $a = 1$ and $b = 1$, the attraction and repulsion shapes of UMAP are given by $f_a^U = -2/(1 + \zeta^2)$ and $f_r^U = 2/(\zeta^2(1 + \zeta^2))$, respectively. Under this setting the attraction shape is bounded to $[-2, 0]$ with $f_a^U(0) = -2$ (Fig. 1(f)), whereas the repulsion shape is essentially unaltered and remains unbounded as $\zeta \rightarrow 0$ (Fig. 1(g)). Because $f_a^U < -1$ as $\zeta \rightarrow 0$, Proposition 4.1 together with the analysis of Section 5 implies that even this unity case requires learning-rate annealing.

Damrich et al. (2023) analyzed UMAP’s negative-sampling loss within the contrastive-embedding (CE) framework and identified the effective UMAP kernel as $1/\zeta^2$. Within CE, the authors defined NEG- t -SNE by replacing this kernel with $1/(1 + \zeta^2)$. Translated back into the UMAP formalism, this corresponds to the low-dimensional affinity $q_{ij}^N = 1/(2 + \zeta^2)$, from which the attraction and repulsion shapes follow as $f_a^N = -2/(2 + \zeta^2)$ and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$, respectively (Figs. 1(f,g)). The attraction shape is now bounded to $[-1, 0]$ and satisfies Proposition 4.1 so any $\lambda \in [0, 1]$ guarantees contraction and precludes oscillation between expansion and contraction. NEG- t -SNE is therefore insensitive to learning-rate annealing, and its clusters remain well resolved even at constant $\lambda = 1.0$. Its repulsion shape is likewise bounded, to $[0, 1]$, and does not diverge as $\zeta \rightarrow 0$, so fewer points are ejected from clusters when sampled as negatives. Whereas Damrich et al. (2023) attributed this disparity solely to the bounded repulsion of NEG- t -SNE relative to the unbounded repulsion of UMAP, our analysis in Fig. 1 demonstrates that the attraction shape contributes equally to cluster stability and compactness. Further discussion and illustrations of NEG- t -SNE are given in Appendix F. Together with the results in Fig. 4 this indicates that attraction and repulsion shapes may be combined across methods, for instance pairing UMAP’s attraction shape with NEG- t -SNE’s repulsion shape. We present such combinations for UMAP, NEG- t -SNE, and PaCMAP in Appendix J.

7 Discussion and Conclusion

In this work we characterized the relationship between the attractive and repulsive forces and cluster formation. Although it is well established that attractive forces contract the distances of similar points and repulsive forces expand those of dissimilar points, the precise mechanisms governing these forces had not been analyzed. By decomposing each force into a distance-dependent shape, we made these mechanisms explicit. While the main text concentrates on UMAP and NEG- t -SNE, the analysis generalizes to dimensionality reduction more broadly (additional algorithms are treated in Appendix G).

Treating the attraction and repulsion coefficients as functions of the pairwise distance and characterizing their shapes (Propositions 4.1 and 4.2) yielded a counterintuitive result: for small distances UMAP’s attractive force produces expansion rather than contraction (Fig. 1(b)), pushing neighboring points apart instead of drawing them together. We showed that UMAP’s learning-rate schedule is precisely the mechanism that suppresses this effect.

We further analyzed the initialization of the low-dimensional embedding. Similar points initialized far apart experience negligible attraction and fail to contract. Our formalism admits a direct remedy: augmenting the attraction shape with additive terms strengthens attraction at large distances. This modification produced embeddings that are substantially more consistent under random initialization, corroborating the conclusion that UMAP encodes a unique underlying structure while the standard numerical devices merely accelerate its recovery.

Analysis of the repulsion shape established that stronger repulsion increases the inter-cluster distance. Motivated by this, we added a small positive constant to the shape, attaining a level of repulsion inaccessible through the conventional route of varying the minimum distance m_d .

Finally, we examined the relationship between NEG- t -SNE and UMAP. In the particular case $a = b = 1$, the essential distinction is that NEG- t -SNE, in its default form, has bounded attraction ($\in [-1, 0]$) and repulsion ($\in [0, 1]$) shapes, so both propositions hold for every learning rate; consequently the method is insensitive to a fixed learning rate in $[0, 1]$. Varying a and b in NEG- t -SNE, however, reintroduces the same numerical difficulties as UMAP. We additionally established that NEG- t -SNE and Parametric UMAP are formally equivalent.

These characterizations of attraction–repulsion dynamics provide concrete tools for tuning dimensionality reduction algorithms. Moreover, given the close correspondence between dimensionality reduction and contrastive learning (Damrich and Hamprecht 2021; Hu et al. 2023), our approach may also inform representation learning. Collectively, this work aims to render embeddings and their interpretation more principled, consistent, and reliable, and to guide subsequent research.

Software and Data

All datasets used in this paper are publicly available. The MNIST dataset is available at yann.lecun.com. Fashion-MNIST is available at github.com.

Single-cell transcriptomes data is available at github.com

References

- Akshay Agrawal, Alnur Ali, Stephen Boyd, et al. Minimum-distortion embedding. *Foundations and Trends® in Machine Learning*, 14(3):211–378, 2021.
- Mahsun Altin and Altan Cakir. Exploring the influence of dimensionality reduction on anomaly detection performance in multivariate time series. *IEEE Access*, 2024.
- Ehsan Amid and Manfred K Warmuth. TriMap: Large-scale dimensionality reduction using triplets. *arXiv preprint arXiv:1910.00204*, 2019.
- Sanjeev Arora, Wei Hu, and Pravesh K Kothari. An analysis of the t-sne algorithm for data visualization. In *Conference On Learning Theory*, pages 1455–1462. PMLR, 2018.
- Etienne Becht, Leland McInnes, John Healy, Charles-Antoine Dutertre, Immanuel WH Kwok, Lai Guan Ng, Florent Ginhoux, and Evan W Newell. Dimensionality reduction for visualizing single-cell data using umap. *Nature biotechnology*, 37(1):38–44, 2019.
- Mikhail Belkin and Partha Niyogi. Laplacian eigenmaps and spectral techniques for embedding and clustering. In *Advances in neural information processing systems*, pages 585–591, 2002.
- Jan Niklas Böhm, Philipp Berens, and Dmitry Kobak. Attraction-repulsion spectrum in neighbor embeddings. *The Journal of Machine Learning Research*, 23(1):4118–4149, 2022.
- Ingwer Borg and Patrick JF Groenen. *Modern multidimensional scaling: Theory and applications*. Springer Science & Business Media, 2007.
- T Tony Cai and Rong Ma. Theoretical foundations of t-sne for visualizing high-dimensional clustered data. *Journal of Machine Learning Research*, 23(301):1–54, 2022.
- Tara Chari and Lior Pachter. The specious art of single-cell genomics. *PLOS Computational Biology*, 19(8):e1011288, 2023.
- Sebastian Damrich and Fred A Hamprecht. On UMAP’s true loss function. *Advances in Neural Information Processing Systems*, 34:5798–5809, 2021.
- Sebastian Damrich, Niklas Böhm, Fred A Hamprecht, and Dmitry Kobak. From t-SNE to UMAP with contrastive learning. In *The Eleventh International Conference on Learning Representations*, 2023.
- Cyril de Bodt, Alex Diaz-Papkovich, Michael Bleher, Kerstin Bunte, Corinna Coupette, Sebastian Damrich, Enrique Fita Sanmartin, Fred A Hamprecht, Emőke-Ágnes Horvát, Dhruv Kohli, et al. Low-dimensional embeddings of high-dimensional data. *arXiv preprint arXiv:2508.15929*, 2025.
- Andrew Draganov and Simon Dohn. Relating tsne and umap to classical dimensionality reduction. *arXiv e-prints*, pages arXiv–2306, 2023.
- Andrew Draganov, Jakob Jørgensen, Katrine Scheel, Davide Mottin, Ira Assent, Tyrus Berry, and Cigdem Aslay. ActUp: Analyzing and consolidating tSNE and UMAP. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI-23*, pages 3651–3658, 8 2023.
- Shu-Kai S Fan, Du-Ming Tsai, Chih-Hung Jen, Chia-Yu Hsu, Fei He, and Li-Ting Juan. Data visualization of anomaly detection in semiconductor processing tools. *IEEE Transactions on Semiconductor Manufacturing*, 35(2):186–197, 2021.
- Jason Fleischer and Mohammad Tariqul Islam. Late breaking abstract-identifying and phenotyping COVID-19 patients using machine learning on chest x-rays. *European Respiratory Journal*, 2020.
- Jonathan Foote. Visualizing music and audio using self-similarity. In *Proceedings of the seventh ACM international conference on Multimedia (Part 1)*, pages 77–80, 1999.
- Rita González-Márquez, Luca Schmidt, Benjamin M Schmidt, Philipp Berens, and Dmitry Kobak. The landscape of biomedical research. *Patterns*, 2024.
- John C Gower. Generalized procrustes analysis. *Psychometrika*, 40:33–51, 1975.
- Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare (HEALTH)*, 3(1):1–23, 2021.

- Geoffrey Hinton and Sam T Roweis. Stochastic neighbor embedding. In *Advances in Neural Information Processing Systems*, volume 15, pages 833–840, 2002.
- Tianyang Hu, Zhili Liu, Fengwei Zhou, Wenjia Wang, and Weiran Huang. Your contrastive learning is secretly doing stochastic neighbor embedding. In *The Eleventh International Conference on Learning Representations*, 2023.
- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning*, pages 20617–20642, 2024.
- Mohammad Tariqul Islam and Jason W Fleischer. Manifold-aligned neighbor embedding. In *ICLR 2022 Workshop on Geometrical and Topological Representation Learning*, 2022.
- Mohammad Tariqul Islam and Jason W Fleischer. Outlier detection in large radiological datasets using umap. In *International Workshop on Topology-and Graph-Informed Imaging Informatics*, pages 111–121. Springer, 2024.
- Robert A Jacobs. Increased rates of convergence through learning rate adaptation. *Neural networks*, 1(4):295–307, 1988.
- Mathieu Jacomy, Tommaso Venturini, Sebastien Heymann, and Mathieu Bastian. Forceatlas2, a continuous graph layout algorithm for handy network visualization designed for the gephi software. *PloS one*, 9(6):e98679, 2014.
- Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.
- Dmitry Kobak and George C Linderman. Initialization is critical for preserving global data structure in both t-sne and umap. *Nature biotechnology*, 39(2):156–157, 2021.
- Dmitry Kobak, George Linderman, Stefan Steinerberger, Yuval Kluger, and Philipp Berens. Heavy-tailed kernels reveal a finer cluster structure in t-sne visualisations. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 124–139. Springer, 2019.
- Alexander Kolpakov and Aidan Rocke. The information geometry of umap. *Le Matematiche*, 79(1):151–164, 2024.
- Nikita Kotlov, Kirill Shaposhnikov, Cagdas Tazearslan, Madison Chasse, Artur Baisangurov, Svetlana Podsvirova, Dawn Fernandez, Mary Abdou, Leznath Kaneunyenye, Kelley Morgan, et al. Procrustes is a machine-learning approach that removes cross-platform batch effects from clinical rna sequencing data. *Communications Biology*, 7(1):392, 2024.
- Joseph B Kruskal. Nonmetric multidimensional scaling: a numerical method. *Psychometrika*, 29(2):115–129, 1964.
- Noël Kury, Dmitry Kobak, and Sebastian Damrich. DREAMS: Preserving both local and global structure in dimensionality reduction. *arXiv preprint arXiv:2508.13747*, 2025.
- Siu Kwan Lam, Antoine Pitrou, and Stanley Seibert. Numba: A llvm-based python jit compiler. In *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC*, pages 1–6, 2015.
- Pierre Lambert, Cyril De Bodt, Michel Verleysen, and John A Lee. Squadmds: a lean stochastic quartet mds improving global structure preservation in neighbor embedding like t-sne and umap. *Neurocomputing*, 503:17–27, 2022.
- George C Linderman and Stefan Steinerberger. Clustering with t-SNE, provably. *SIAM Journal on Mathematics of Data Science*, 1(2):313–332, 2019.
- George C Linderman and Stefan Steinerberger. Dimensionality reduction via dynamical systems: The case of t-sne. *SIAM Review*, 64(1):153–178, 2022.
- Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of machine learning research*, 9(Nov):2579–2605, 2008.
- Evan Z Macosko, Anindita Basu, Rahul Satija, James Nemesh, Karthik Shekhar, Melissa Goldman, Itay Tirosh, Allison R Bialas, Nolan Kamitaki, Emily M Martersteck, et al. Highly parallel genome-wide expression profiling of individual cells using nanoliter droplets. *Cell*, 161(5):1202–1214, 2015.
- Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018.
- Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. Distributed representations of words and phrases and their compositionality. In *Advances in neural information processing systems*, pages 3111–3119, 2013.
- Ashwin Narayan, Bonnie Berger, and Hyunghoon Cho. Assessing single-cell transcriptomic variability through density-preserving data visualization. *Nature Biotechnology*, 39(6):765–774, 2021.
- F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

- Pavlin G. Poličar, Martin Stražar, and Blaž Zupan. `opentsne`: A modular python library for t-sne dimensionality reduction and embedding. *Journal of Statistical Software*, 109(3):1–30, 2024.
- Aditya Ravuri and Neil D Lawrence. Towards one model for classical dimensionality reduction: A probabilistic perspective on umap and t-sne. *arXiv preprint arXiv:2405.17412*, 2024.
- Aditya Ravuri, Francisco Vargas, Vidhi Lalchand, and Neil D Lawrence. Dimensionality reduction as probabilistic inference. In *ICML 2023 Workshop on Structured Probabilistic Inference & Generative Modeling*, 2023.
- Peter J Rousseeuw. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of computational and applied mathematics*, 20:53–65, 1987.
- Sam T Roweis and Lawrence K Saul. Nonlinear dimensionality reduction by locally linear embedding. *science*, 290(5500):2323–2326, 2000.
- Tim Sainburg, Leland McInnes, and Timothy Q Gentner. Parametric umap embeddings for representation and semisupervised learning. *Neural Computation*, 33(11):2881–2907, 2021.
- Uri Shaham and Stefan Steinerberger. Stochastic neighbor embedding separates well-separated clusters. *arXiv preprint arXiv:1702.02670*, 2017.
- Xintong Shi, Wenzhi Cao, and Sebastian Raschka. Deep neural networks for rank-consistent ordinal regression based on conditional probabilities. *Pattern Analysis and Applications*, 26(3):941–955, 2023.
- Jian Tang, Jingzhou Liu, Ming Zhang, and Qiaozhu Mei. Visualizing large-scale and high-dimensional data. In *Proceedings of the 25th international conference on world wide web*, pages 287–297, 2016.
- Laurens Van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of machine learning research*, 9(11), 2008.
- Jarkko Venna and Samuel Kaski. Neighborhood preservation in nonlinear projection methods: An experimental study. In *International Conference on Artificial Neural Networks*, pages 485–491. Springer, 2001.
- Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.
- Yingfan Wang, Haiyang Huang, Cynthia Rudin, and Yaron Shaposhnik. Understanding how dimension reduction tools work: an empirical approach to deciphering t-SNE, UMAP, TriMAP, and PaCMAP for data visualization. *The Journal of Machine Learning Research*, 22(1):9129–9201, 2021.
- Yingfan Wang, Yiyang Sun, Haiyang Huang, and Cynthia Rudin. Dimension reduction with locally adjusted graphs. In *Thirty-Ninth AAAI Conference on Artificial Intelligence*, 2025.
- Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. 2017.
- Yang Yang, Hongjian Sun, Jialei Gong, Yali Du, and Di Yu. Interpretable dimensionality reduction by feature preserving manifold approximation and projection. *arXiv preprint arXiv:2211.09321*, 2022.
- Lihi Zelnik-Manor and Pietro Perona. Self-tuning spectral clustering. *Advances in neural information processing systems*, 17, 2004.
- Jonathan X Zheng, Samraat Pawar, and Dan FM Goodman. Graph drawing by stochastic gradient descent. *IEEE transactions on visualization and computer graphics*, 25(9):2738–2748, 2018.

Appendix

Appendix [A](#) provides the derivations of the attraction and repulsion shapes of UMAP. Extending the main-text analysis, Appendix [B](#) further examines the constant-learning-rate regime, and Appendix [C](#) discusses the implications of strengthened attraction and repulsion for large language model embeddings. Appendix [D.1](#) formally defines the Procrustes distance and the Procrustes matrix. Appendix [E](#) reports the random-initialization experiment on additional datasets, and Appendix [F](#) extends the comparison between NEG- t -SNE and UMAP. Appendix [G](#) analyzes alternative dimensionality reduction methods, and Appendix [H](#) details the construction of the high-dimensional graph in these methods. Appendix [I](#) presents the full results of the λ_a - λ_r sweep (Fig. [1](#)) for UMAP, NEG- t -SNE, and PaCMAP across datasets, and Appendix [J](#) reports the effect of combining shapes across algorithms. Finally, Appendix [K](#) provides the implementation details.

A Proofs and Derivations

A.1 Proof of Proposition [4.1](#)

Proof. We subtract Eq. [\(7\)](#) from Eq. [\(8\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

This distance contracts as long as $|1 + 2\lambda f_a| < 1$, i.e., provided

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition [4.2](#)

Proof. We subtract Eq. [\(9\)](#) from Eq. [\(10\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

This distance increases when $|1 + \lambda f_r| > 1$, i.e.,

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

From the definition of f_r , the latter suffices. □

A.3 Attraction Shape:

To derive the attraction shape we adopt a general form of the low-dimensional affinity, $q_{ij} = (\gamma + a\|y_i - y_j\|_2^{2b})^{-1}$, which recovers UMAP for $\gamma = 1$ and NEG- t -SNE for $\gamma = 2$. The attractive force is then

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

Defining $\zeta = \|y_i - y_j\|_2$, the scalar prefactor is the attraction shape:

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

A.4 Condition for strictly increasing f_a^U :

The monotonicity of f_a^U in ζ is determined by its derivative:

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

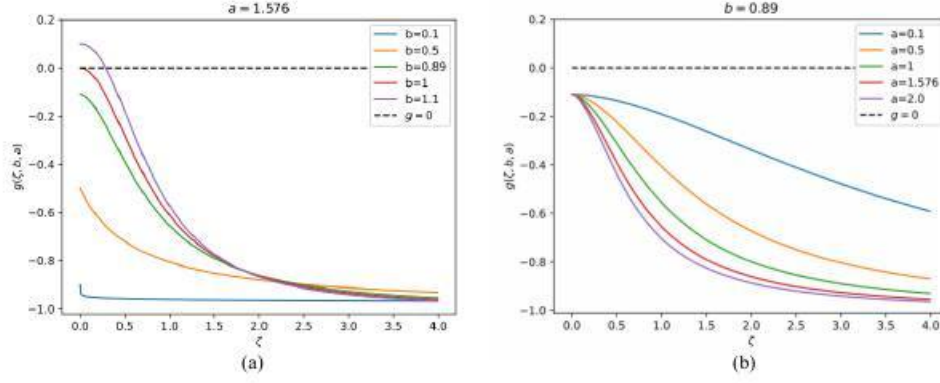


Figure 5: Values of $g(\zeta, b, a)$ (a) with a fixed at 1.576 and (b) with b fixed at 0.89.

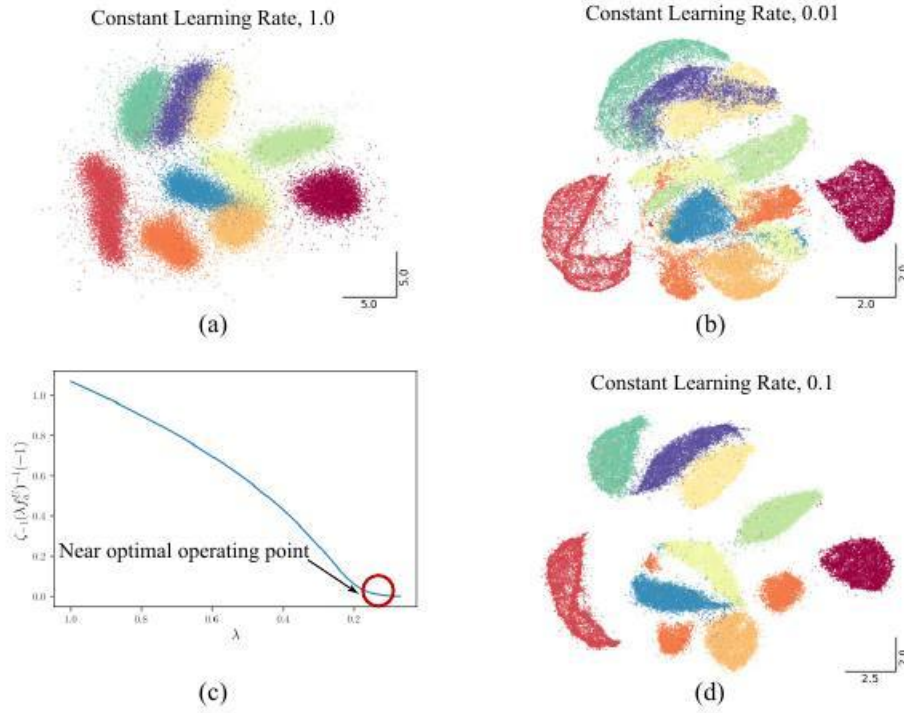


Figure 6: Effect of a constant learning rate on the embeddings. (a) At a large learning rate ($\lambda = 1.0$), the embedding is diffuse, owing to the large value of ζ_{-1} . (b) At a small learning rate ($\lambda = 0.01$), clusters fail to form, as the attractive and repulsive forces are too weak. (c) ζ_{-1} decreases nonlinearly with the learning rate; the objective is to reduce ζ_{-1} while retaining effective attraction and repulsion. (d) Distinct, compact clusters form at the constant near-optimal learning rate $\lambda = 0.1$.

This leads to a strictly increasing condition ($\frac{df^U}{d\zeta} > 0$),

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. Examining the derivative and asymptotes of g shows that this inequality holds for all ζ precisely when $0 < b \leq 1$. Figure 5 plots g for a range of b and a ; once b exceeds 1, the inequality (23) is violated.

A.5 Repulsion Shape:

Using the same general form of the low-dimensional affinity, the repulsive force is

$$\begin{aligned}
 \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\
 &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j).
 \end{aligned} \tag{24}$$

The scalar prefactor is the repulsion shape:

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}. \tag{25}$$

Generally, $f_r > 0$.

A.6 Loss functions due to modified attraction and repulsion shape:

Integrating the modified attractive force of Eq. (13) recovers the corresponding cost function,

$$- \int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j) dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \tag{26}$$

and integrating the modified repulsive force of Eq. (15) yields

$$- \int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j) dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \tag{27}$$

In both cases the additional term is a norm-based regularizer of the loss. Adding the term directly to the shapes, however, exposes its role transparently: in the attractive case it increases far-sightedness, and in the repulsive case it imposes a constant additive repulsive coefficient.

B More on UMAP’s Learning Rate

As discussed in Sections 5 and 6 UMAP is widely held to require learning-rate annealing (Fig 6). Section 5 accounts for this by introducing the minimum distance for contraction (ζ_{-1}) and establishing that reducing it via annealing yields compact clusters. Section 6 compared the attraction shapes of UMAP (at $a = b = 1.0$) and Neg- t -SNE and showed that Neg- t -SNE tolerates a constant learning rate of 1.0 better than UMAP because its attraction shape lies in $[-1, 0]$ whereas UMAP’s lies in $[-2, 0]$. By the same argument, UMAP tolerates a constant learning rate of 0.5, since this confines its effective attraction shape to $[-1, 0]$ (Figs. 11(f,h)).

Even so, annealing the learning rate produces better embeddings across a wide range of a and b . The reason is that the algorithm’s ultimate objective is $\zeta_{-1} \rightarrow 0$, which is attained only as the learning rate approaches zero; otherwise the embedding remains diffuse (Fig 6(a)). Conversely, if the learning rate is initialized too low, the attractive and repulsive forces are too weak for any clear cluster to form (Fig 6(b)). The ζ_{-1} -versus- λ curve (Fig 6(c)) identifies $\lambda = 0.1$ as a near-optimal operating point, at which ζ_{-1} is small while attraction and repulsion remain substantial. Fixing the constant learning rate at 0.1 accordingly yields compact clusters with sharp boundaries (Fig 6(d)).

C Large Language Model Embedding

Strengthening attraction and repulsion frequently improves the resulting embedding. To demonstrate this, we drew a random sample of 300,000 PubMedBERT (Gu et al. 2021) feature vectors with known labels from the PubMed dataset (González-Márquez et al. 2024). Default UMAP yields a crowded layout in which the label colors overlap (Fig. 7(a)); absent the labels, the embedding is difficult to interpret because no explicit clusters are present. Reducing b to 0.6 (per Section 5.2) increases repulsion while simultaneously promoting cluster formation through strengthened attraction, producing a layout that is interpretable even without labels (Fig. 7(b)). The improvement is quantified by the k -NN classification accuracy ($k = 10$): on a held-out set of 10,000 points, reducing b to 0.6 raises the accuracy from 49.2% to 55.48%.

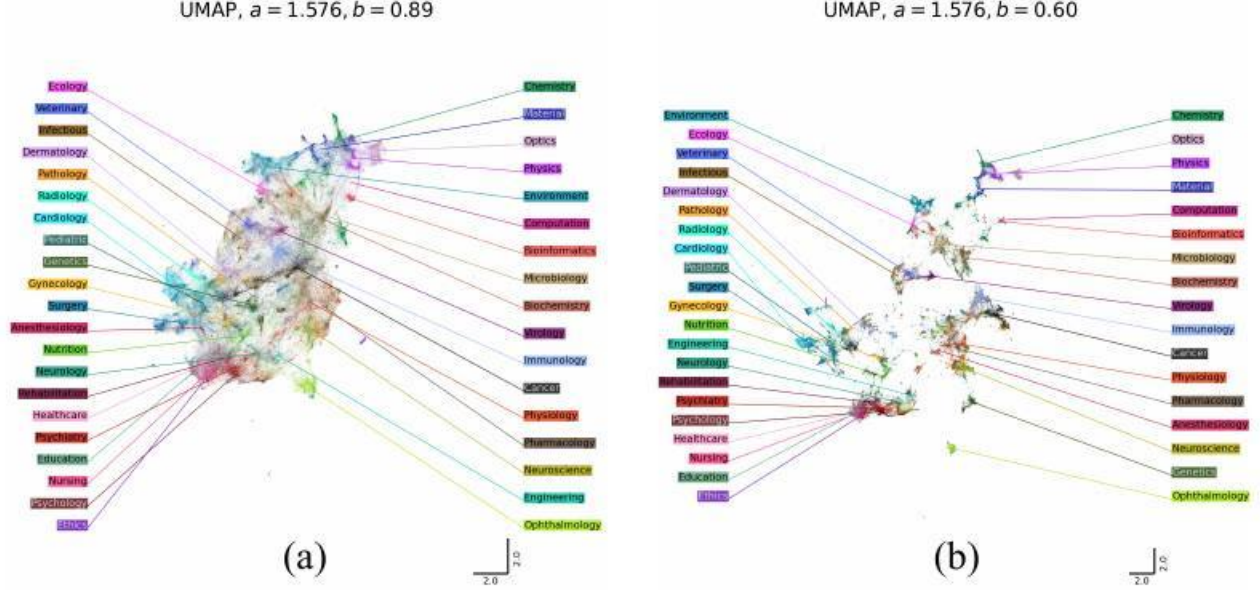


Figure 7: Visualization of 300K abstracts from the PubMed dataset. (a) Default UMAP ($a = 1.576, b = 0.89$) produces overlapping labels in the mapping, whereas (b) UMAP with strengthened attraction and repulsion ($a = 1.576, b = 0.60$) produces well-separated clusters with non-overlapping labels. Both mappings are annotated with labels to aid interpretation.

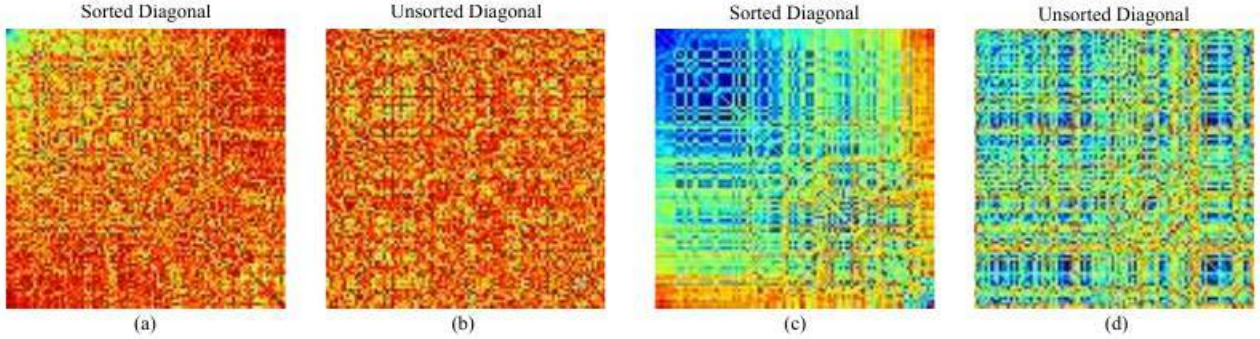


Figure 8: Effect of sorting the diagonal of the Procrustes matrix on its visualization. (a) Procrustes matrix reproduced from Fig. 2(f) for the default UMAP attraction shape, with the diagonal sorted by p_d to a reference PCA-initialized embedding. (b) The same data as in (a) with the diagonal left unsorted (equivalently, randomly ordered). (c) Procrustes matrix reproduced from Fig. 2(g) for the modified attraction shape, with the diagonal sorted by p_d to the reference PCA-initialized embedding. (d) The same data as in (c) with the diagonal left unsorted.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

The Procrustes distance (Gower [1975]) quantifies the dissimilarity between two point clouds $\{x\}$ and $\{y\}$ modulo the linear transformations of translation, scaling, and rotation. Operationally, $\{x\}$ is held fixed and $\{y\}$ is transformed until the two sets attain maximum alignment. Denoting by $\{y'\}$ the optimally aligned transform of $\{y\}$, the Procrustes distance is

$$p_d(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

The Procrustes distance is a linear similarity measure that has been applied across a range of settings (McInnes et al. [2018], Islam and Fleischer [2022], Kotlov et al. [2024]).

We use the Procrustes distance to assess the consistency of the embedding under random initialization. Let $\{x\}_p$ denote a reference embedding (obtained with PCA initialization in our experiments), and let $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ denote the set of N embeddings obtained under random initialization. The mutual consistency of the embeddings is summarized by the mean and standard deviation of the strictly lower-triangular entries of the matrix P (reported as mean $\pm$ std in Figs. 2, 9 and 10):

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

The indices of X_r are ordered so that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$; the diagonal entries of the Procrustes matrix are then

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal entries are

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Sorting the diagonal has no numerical effect on the summary statistics, but it renders the self-similarity of the embeddings visually apparent. In Fig. 8(a), for example, the sorted diagonal concentrates the mutually similar embeddings in the upper-left region of the matrix. Under the modified attraction shape, the number of mutually similar embeddings increases, as reflected by the enlarged blue region in Fig. 8(c), which indicates a metastable configuration of the embedding algorithm. When the diagonal is unsorted, this region is not resolved and the self-similarity structure is no longer discernible (Fig. 8(b,d)).

D.2 Trustworthiness

The trustworthiness metric (Venna and Kaski 2001) quantifies the extent to which local neighborhoods are preserved under dimensionality reduction:

$$T = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ denotes the k -NN graph in the embedding space and $r(i, j)$ denotes the rank of x_j in the high-dimensional k -NN graph. We set $k = 5$, so the metric assesses preservation of each point's five nearest neighbors. For computational efficiency, trustworthiness is evaluated over a random sample of 10,000 indices, and the same indices are used across all embeddings being compared.

D.3 Silhouette Score

Although the silhouette score (Rousseeuw 1987) is conventionally used to evaluate clustering algorithms, we repurpose it to measure label separation in the embedding, i.e., how well the ground-truth labels are separated into clusters; the underlying premise is that the embedding algorithm should partition the labels as distinctly as possible. For a point y_i in a point cloud $\{y\}$, let a_i denote the mean distance from y_i to the remaining points sharing its label and b_i the minimum, over all other labels, of the mean distance from y_i to the points of that label. The pointwise silhouette is

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This value lies in $[-1, 1]$: values near 1 indicate that y_i is well embedded within its own label cluster, values near 0 indicate a boundary point, and values near -1 indicate failed label separation. The overall silhouette score is the mean over all points,

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

The silhouette score is computed over the entire embedding, without random subsampling, using Euclidean distances.

E Effect of Random Initialization in Additional Datasets

The main text (Section 5.1) reports results on MNIST alone. Here we replicate the identical protocol on the Fashion-MNIST (FMNIST) (Xiao et al. 2017) and single-cell transcriptomes (Macosko et al. 2015) datasets (Figs. 9 and 10 respectively). The principal conclusion is unchanged: the modified and composite attraction shapes, which strengthen attraction at large distances, substantially improve the consistency of the reconstruction.

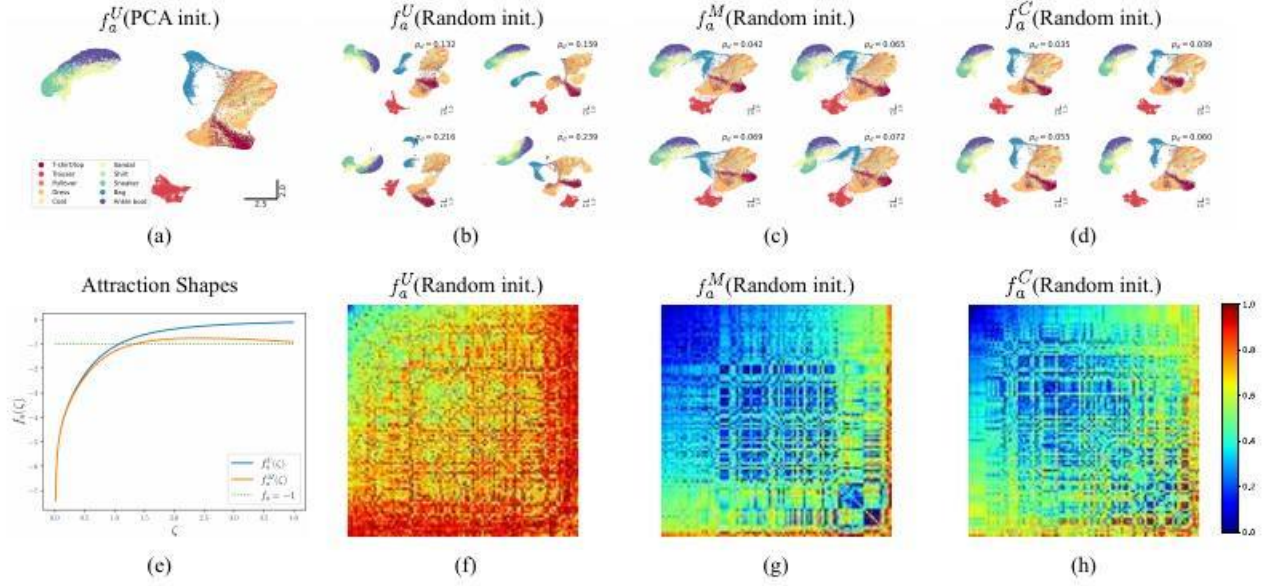


Figure 9: Effect of random UMAP initialization on distinct attraction shapes for the FMNIST dataset. (a) Reference mapping obtained with PCA initialization. (b-d) The four mappings attaining the lowest Procrustes distance (p_d) to the reference in (a), for the (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices aggregated over 100 runs for the (f) default (0.72 ± 0.15), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18) attraction shapes.

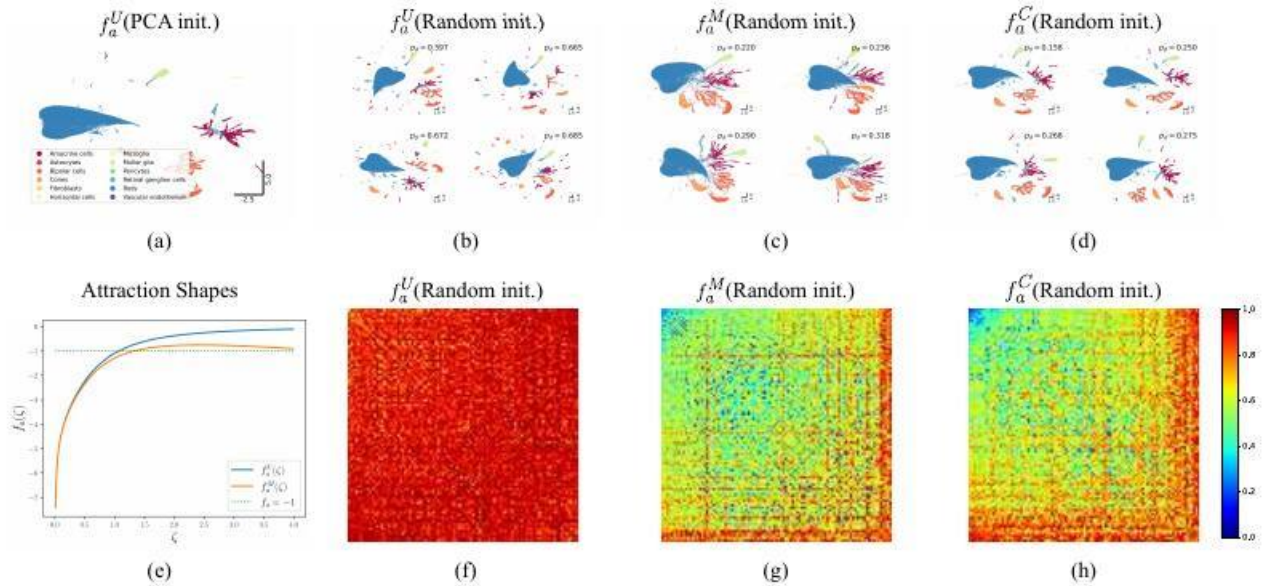


Figure 10: Effect of random UMAP initialization on distinct attraction shapes for the single-cell transcriptomes dataset. (a) Reference mapping obtained with PCA initialization. (b-d) The four mappings attaining the lowest Procrustes distance (p_d) to the reference in (a), for the (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices aggregated over 100 runs for the (f) default (0.91 ± 0.06), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15) attraction shapes.

F More on NEG- t -SNE

F.1 Comparison to UMAP

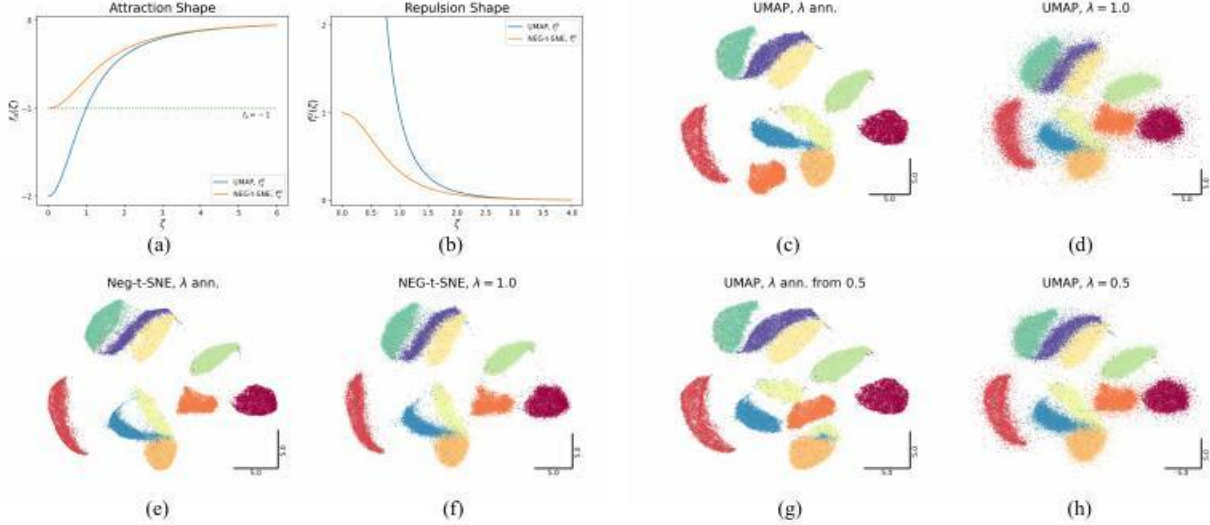


Figure 11: Sensitivity of UMAP and NEG- t -SNE to the learning rate on the MNIST dataset. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1$, $b = 1$) and NEG- t -SNE. (c,d) UMAP is highly sensitive to the learning rate λ , since $f_a^U < -1$ as the separation ζ decreases; without annealing, the clusters become diffuse. (e,f) NEG- t -SNE is comparatively insensitive to λ , since $f_a^N \in [-1, 0]$ for all ζ , so its clusters remain well resolved even without annealing. (g,h) Confining f_a^U to $[-1, 0]$ by setting $\lambda = 0.5$ likewise reduces the sensitivity to λ .

Figure 11 shows the shapes of UMAP and NEG- t -SNE together with the corresponding MNIST embeddings. Under an annealed learning rate, the two methods yield comparable outputs (Figs. 11(c,e)). At a constant learning rate of 1, however, UMAP produces a diffuse structure whereas NEG- t -SNE produces markedly sharper boundaries (Fig. 11(d,f)). Section 6 predicts that confining f_a^U to $[-1, 0]$ should yield less diffuse clusters at fixed λ . This is consistent with Fig. 11(h,i), where UMAP attained better embedding and clustering when the attraction learning rate satisfied $\lambda_a \lesssim 0.5$. A direct way to enforce the constraint is to set $\lambda = 0.5$, which satisfies Proposition 4.1 for all ζ . The resulting embeddings, in Figs. 11(g) and (h), confirm that the clusters resemble those of NEG- t -SNE and that, at constant $\lambda = 0.5$, they exhibit sharper boundaries than before, as predicted (Fig. 11(d,h)). A small number of points nonetheless remain outside the clusters, a consequence of UMAP’s repulsion shape that NEG- t -SNE resolves.

We now introduce the parameters a and b into NEG- t -SNE, which recovers the Parametric UMAP formulation (Proposition 6.1). The affinity becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, and the attraction and repulsion shapes become $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2+a\zeta^{2b}}$ and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1+a\zeta^{2b})(2+a\zeta^{2b})}$, respectively. For $0 < b < 1$, both shapes diverge as $\zeta \rightarrow 0$, so NEG- t -SNE inherits the same numerical difficulties as UMAP whenever a and b are varied. One notable distinction is that its attraction shape attains a smaller minimum distance ζ_{-1} ; while this may enhance cluster formation, the shape also decays to zero more rapidly (heightened near-sightedness at increasing distance), which can diminish its effectiveness for attraction over long distances.

F.2 Comparison to Parametric UMAP

Parametric UMAP was originally trained with the original UMAP objective (Sainburg et al., 2021), but subsequent work adopted a numerically stable, log-sigmoid-based modified cross-entropy loss (Shi et al., 2023). Under this modification, Parametric UMAP and Neg- t -SNE (Danrich et al., 2023) coincide (Proposition 6.1), as we establish below.

Proof of Proposition 6.1

Proof. Under this modification, the kernel function is

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

And the repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Both terms coincide with those of NEG- t -SNE carrying explicit parameters a and b (which reduce to 1 in the default NEG- t -SNE). $\square$

G Alternate Dimensionality Reduction Algorithms

The alternative algorithms considered here share UMAP's kernel function, specialized to $a = b = 1$ in the low-dimensional weight:

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

We first analyze the TriMap (Amid and Warmuth, 2019) algorithm. Although TriMap operates on triplets rather than pairwise interactions, it serves as an instructive case in which the attraction and repulsion analysis is somewhat more involved than for UMAP. We then treat Pairwise Controlled Manifold Approximation (PaCMAP) (Wang et al., 2021) and its extension, Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP) (Wang et al., 2025), which recast TriMap's loss for pairwise interactions. Next we analyze t -SNE (Van der Maaten and Hinton, 2008), provide a brief treatment of SNE (Hinton and Roweis, 2002), which employs an alternative kernel, and close the section with multidimensional scaling (Borg and Groenen, 2007).

G.1 TriMap

TriMap (Amid and Warmuth, 2019) optimizes the low-dimensional embedding through a triplet loss

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} is the weight of the triplet (y_i, y_j, y_k) , y_j lies in the high-dimensional k -nearest-neighbor set of y_i , and y_k is a distant point. Minimizing this loss drives y_i and y_j to attract and y_i and y_k to repel. The corresponding update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

where $\zeta_1 = \|y_i - y_j\|_2$ is the nearest-neighbor distance, $\zeta_2 = \|y_i - y_k\|_2$ is the distance to the distant point, and f_a^T and f_r^T are the attraction and repulsion shapes of TriMap, respectively. In contrast to UMAP, the attractive and repulsive components are non-separable, and each shape depends on the two distances ζ_1 and ζ_2 (i.e., the shapes are two-dimensional). The attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

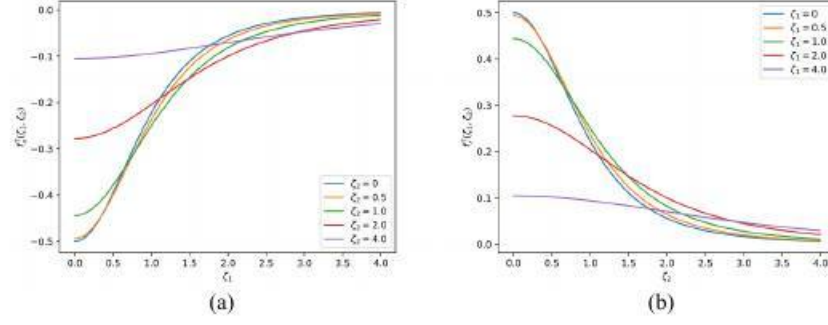


Figure 12: (a) Attraction shapes of TriMap for a range of ζ_2 and (b) repulsion shapes of TriMap for a range of ζ_1 .

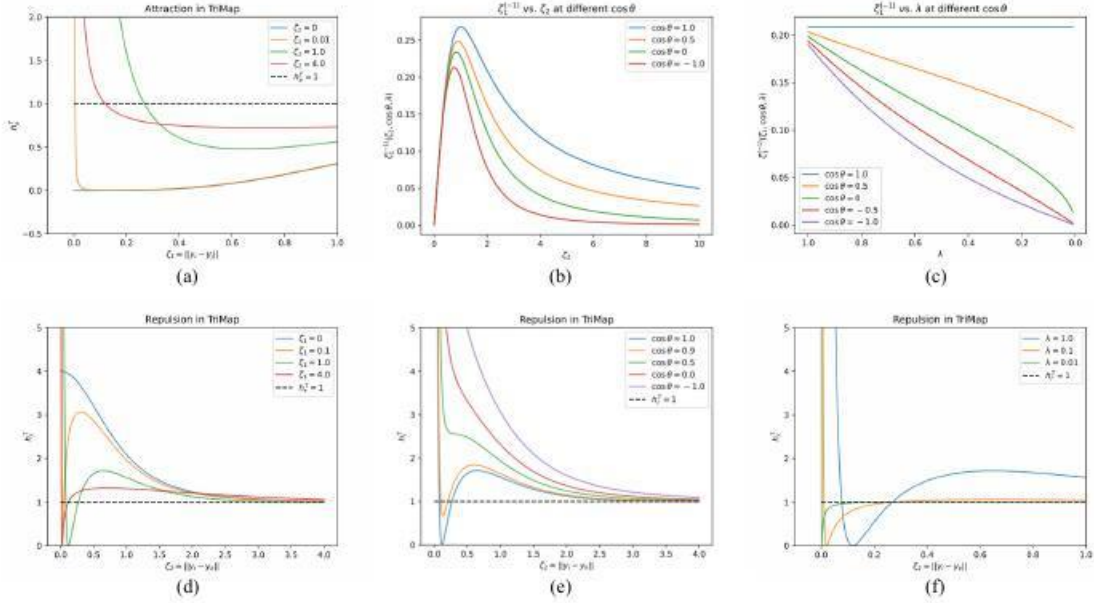


Figure 13: Attraction and repulsion behavior in TriMap. (a) h_a^T versus ζ_1 for a range of ζ_2 ; values below the dotted line indicate attraction. As in UMAP, TriMap exhibits both attraction and repulsion for the nearest-neighbor pair (y_i, y_j) . (b,c) In contrast to UMAP, the minimum distance for contraction ($\zeta_1^{(-1)}$) varies with (b) ζ_2 and (c) λ , with $\cos \theta$ setting the range of admissible values. (d-f) Repulsion in TriMap as a function of (d) ζ_1 , (e) $\cos \theta$, and (f) λ ; values above (below) the dotted line indicate repulsion (attraction). Whereas UMAP's repulsive force yields only repulsion, TriMap's can yield both attraction and repulsion. Unless otherwise indicated, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

The attraction and repulsion shapes of TriMap (Fig. 12) follow trends analogous to those of UMAP. The attraction shape, however, is bounded below by -0.5 , so, unlike UMAP, no endpoint crossover occurs under attraction alone. Because the updates (45)-(47) do not decouple the attractive and repulsive terms, Propositions 4.1 and 4.2 do not apply directly. Considering attraction first, we obtain the following:

Proposition G.1. *Update equations (45)-(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i - y_j)^T (y_i - y_k)}{\|y_i - y_j\|_2 \|y_i - y_k\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46): $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T(y_i^t - y_k^t)$. Putting this value in Eq. (51), we obtain the desired inequality. $\square$

Inequality (50) depends jointly on ζ_1 , ζ_2 , $\cos \theta$, and λ . In particular, the distance $\zeta_1 = \|y_i - y_j\|_2$ that we wish to contract is coupled to the remaining variables. Figure 13(a) shows the attraction behavior across values of ζ_2 at $\cos \theta = 1$ and $\lambda = 1$: values below the dotted line indicate attraction (contraction of ζ_1), and values above indicate repulsion (expansion of ζ_1). The intersection of h_a^T with the dotted line defines the minimum distance for contraction ($\zeta_1^{(-1)}$), the TriMap analogue of UMAP's ζ_{-1} :

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

In most cases $\zeta_1^{(-1)}$ is finite and strictly positive (Fig. 13(b,c)). Consequently, TriMap can exhibit UMAP-like behavior and thus requires learning-rate annealing or an equivalent mechanism; the TriMap implementation uses the delta-bar-delta method (Jacobs 1988) together with an appropriate initialization.

Proposition G.2. *Update equations (45)-(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T(1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47): $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T(y_i^t - y_j^t)$. Putting this value in Eq. (55) we obtain inequality (54). $\square$

Inequality (54) likewise depends jointly on ζ_1 , ζ_2 , $\cos \theta$, and λ . Here the quantity of interest is the expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d-f) show the repulsion behavior as the remaining quantities are varied: values above the dotted line indicate repulsion (expansion), and values below indicate attraction. In marked contrast to UMAP, repulsion in TriMap can produce contraction rather than expansion. Because this anomaly is confined to small distances, it can be avoided by suitable initialization and triplet selection.

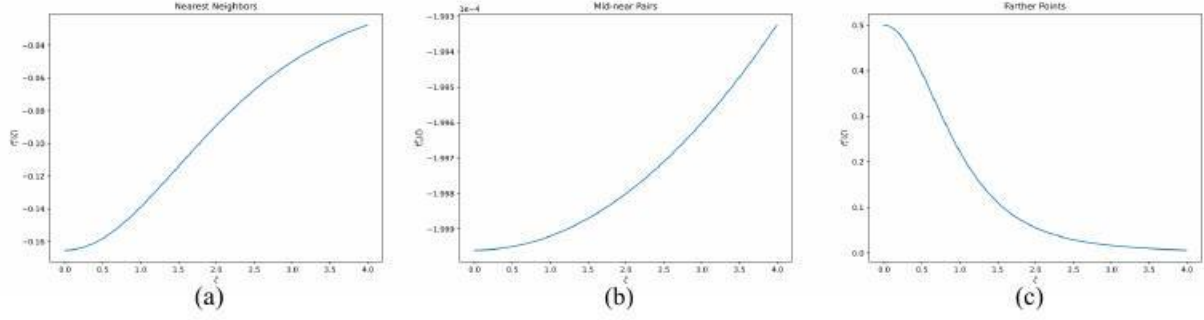


Figure 14: Attraction and repulsion shapes of PaCMAP. (a,b) Attraction shapes for the (a) nearest-neighbor and (b) mid-near pairs (note that the latter values are of order 10^{-4}). (c) Repulsion shape for the farther pairs. All panels use $\lambda = 1$.

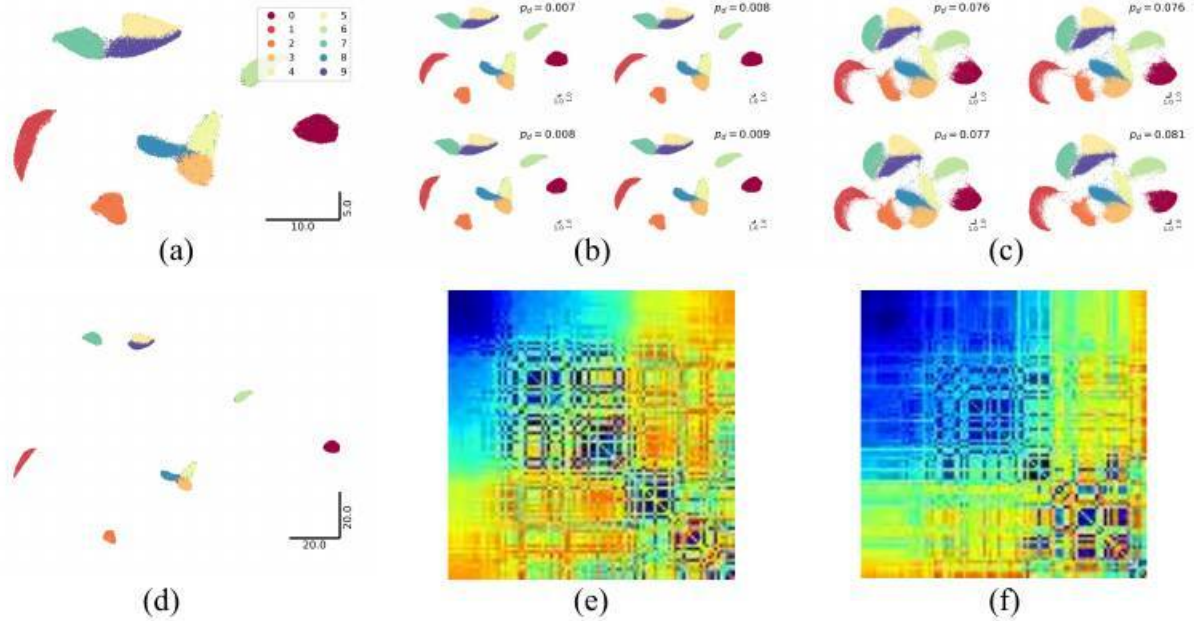


Figure 15: PaCMAP behavior under distinct conditions. (a) PaCMAP embedding of MNIST with PCA initialization. (b,c) The four randomly initialized embeddings best matching (a) for (b) the original PaCMAP and (c) PaCMAP with the modified attraction shape $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$. (d) PaCMAP of MNIST with the modified repulsion shape $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrices for (e) the original PaCMAP (0.51 ± 0.22) and (f) PaCMAP with the modified attraction shape (0.43 ± 0.22). As in UMAP, strengthening attraction at large distances improves consistency, while strengthening repulsion yields smaller clusters and larger inter-cluster distances.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al., 2021) optimizes the low-dimensional embedding across multiple scales. Its loss function is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} are the weights of the nearest-neighbor (NN) pairs, mid-near (MN) pairs, and farther pairs (FP), respectively (details in Appendix H). The first two terms are attractive and the last is repulsive. The loss is a modified form of TriMap’s triplet loss: in the attractive terms, TriMap’s affinity to distant points is replaced by constants ($1/10$ for nearest neighbors, $1/10000$ for mid-near pairs, and 1 for farther points). The loss thus separates

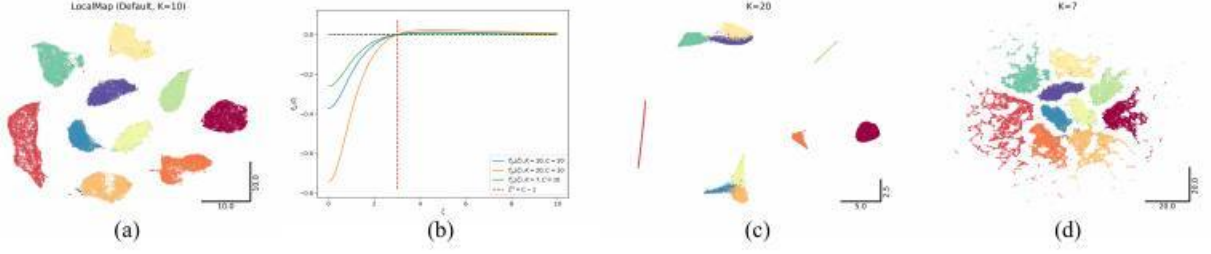


Figure 16: Behavior of LocalMAP on MNIST. (a) Default embedding. (b) Nearest-neighbor attraction–repulsion shape; values below (above) the dotted line indicate attraction (repulsion), with the transition occurring as ζ increases past $\sqrt{C-1}$. Following the LocalMAP implementation, we set $C = 10$. (c) For large $K (= 20)$, repulsion dominates and the clusters become compact. (d) For small $K (= 7)$, attraction dominates and the clusters fragment.

into three terms with decoupled update equations. For the nearest-neighbor term, the updates are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

for the mid-near pairs,

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and for the farthest pairs,

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, and $\zeta_3 = \|y_i - y_l\|_2$ are the respective distances, f_a^P and f_m^P are the attraction shapes for the nearest-neighbor and mid-near pairs, and f_r^P is the repulsion shape for the farthest pairs. The corresponding functional forms are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

Here f_a^P and f_m^P satisfy Proposition 4.1 and f_r^P satisfies Proposition 4.2 (Fig. 14). The attraction is substantially weaker than in UMAP but remains effective across a wide range of learning rates, which are modulated by the Adam optimizer (Kingma and Ba 2015); with $w_{NB} = 3$, the attraction magnitude never exceeds 0.5, precluding endpoint crossover during the attractive update. PaCMAP typically initializes the embedding within a small sphere in the low dimension (on the order of 10^{-3} in (Wang et al. 2021)) and relies on repulsion to separate the clusters. It largely recovers UMAP’s clustering properties, particularly on MNIST, with improved ordering. Its consistency under random initialization exceeds that of UMAP (Fig. 15b,e)) and can be improved further with a modified attraction shape (Fig. 15c,f)). Adding a small constant to the repulsion shape increases the compactness of the embedding by enlarging the inter-cluster distance.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al. 2025) is a refinement of PaCMAP. A defining property of LocalMAP is that it separates all 10 MNIST clusters (Fig. 16a), consistent with the behavior in Figs. 3g,h,j,k) and Figs. 4a,b). Here we analyze how the interplay of attractive and repulsive forces governs the compactness and connectedness of the clusters.

LocalMAP first runs PaCMAP and then performs additional optimization on the attraction to decouple clusters, minimizing the loss

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term combines attractive and repulsive behavior within a single nearest-neighbor pair: in one distance regime it is attractive, and in the other repulsive. The second term is identical to PaCMAP's, save that the farther pairs are resampled every few iterations. We therefore analyze only the first, nearest-neighbor term. Its update equations are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_i^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_i^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the combined attraction–repulsion shape,

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)^2}. \quad (69)$$

By Proposition 4.1 the update contracts the pair whenever $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$. When $\zeta^2 > C-1$, we have $f_{ar}^L > 0$, and by Proposition 4.2 the update expands the pair. The parameters K and C jointly determine whether attraction or repulsion dominates. The LocalMAP implementation sets $C = K = 10$ (Fig. 16(b)), yielding a stronger attraction than PaCMAP (Fig. 14(a)). Consequently, for $\zeta > 3$ the nearest neighbors experience repulsion, so pairs that bridge two clusters separate. The scale parameter K modulates the magnitude of the forces and thereby regulates this separation.

By Proposition 4.1 the condition $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ bounds K , and hence the maximum attraction attainable by f_{ar}^L , without inducing endpoint crossover. (Avoiding crossover during LocalMAP optimization preserves the ordering inherited from PaCMAP, which is necessary for cluster separation.) Substituting into (69) gives $K \leq \frac{\lambda(1+C)^2}{C-1}$, which for $C = 10$ and $\lambda = 1$ yields $K \lesssim 13.44$. For $K \gtrsim 13.44$, the repulsive forces dominate and the clusters become more compact, but the LocalMAP objective is not met, as the inter-cluster bridge persists (Fig. 16(c), $K = 20$). Conversely, as K decreases the attractive forces dominate, because the repulsive forces become too weak, and existing clusters fragment (Fig. 16(d), $K = 7$, mirroring Fig. 4(b)).

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton 2008) optimizes over the pairwise distances. Its loss function is

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the weight of the pair. The original t -SNE implementation considers all pairs, not only nearest neighbors. This loss decomposes into attractive and repulsive contributions as

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As before, the first term is attractive and the second repulsive. The attractive term coincides with that of UMAP at $a = b = 1$ and is inexpensive to evaluate, whereas the repulsive term couples all pairs through the normalization Z and is therefore computationally costly. Applying the same decomposition as for UMAP yields the t -SNE update equations. Because the original t -SNE does not restrict to the nearest-neighbor graph, the weight $w_{i,j}$, evaluated over all pairs, enters the updates directly. The attractive updates are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the t -SNE attraction shape and $\zeta_{i,j} = \|y_i - y_j\|_2$. The repulsive updates are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k}^t)(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j}^t)(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the t -SNE repulsion shape and $Z = \sum_{k \neq l} q_{k,l}$ is the normalization constant. The functional forms are

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1+\zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1+\zeta^2)^2}. \quad (77)$$

From the attractive updates (72)-(73), f_a^{t-SNE} satisfies Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$) and defines a minimum distance for contraction ζ_{-1} . The repulsive update, being coupled across all pairs, admits no simple relation to the repulsion shape; drawing on the TriMap analysis, we instead establish the following:

Proposition G.3. *The update Eqs. (74)-(75) provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k,k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l,l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

Proof. We require,

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Putting this value in Eq. (79), we obtain the desired inequality. $\square$

Because t -SNE's repulsion condition (Eq. 78) is structurally identical to that of TriMap, its repulsion behavior is the same: t -SNE's repulsive forces can produce both attraction and repulsion.

Since t -SNE's forces are scaled, by $w_{i,j}$ for attraction and $w_{i,j}/Z$ for repulsion, both are typically weaker than in UMAP. The algorithm therefore employs large learning rates, e.g., $\approx 10^3$ in the Open- t -SNE package (Poličar et al. 2024). Moreover, most t -SNE implementations include an early-exaggeration phase, in which the attractive forces are multiplied by a constant during the first few iterations. This draws together points that should be proximate but are currently far apart, thereby inducing far-sightedness; conversely, when points are very close ($\zeta_{i,j} \approx \zeta_{-1}$) yet require separation, the same mechanism effects that separation. Early exaggeration is thus essential for obtaining a consistent t -SNE embedding, particularly under random initialization. Applied throughout the optimization rather than only initially, this mechanism renders t -SNE embeddings visually closer to those of UMAP (Böhm et al. 2022).

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis, 2002) is among the earliest algorithms in this class. It shares the t -SNE loss (Eq. 71) but replaces the kernel with the Gaussian $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$, giving the attraction shape

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

that follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$) and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

which satisfies Proposition G.3 under the substitution of the SNE quantities for their t -SNE counterparts. The attraction shape is constant and thus ill-conditioned, so contraction depends entirely on the learning rate λ and the weights $w_{i,j}$; the resulting clusters overlap even for small sample sizes (the crowding problem (Van der Maaten and Hinton, 2008)). t -SNE and later algorithms mitigate this by replacing the Gaussian kernel with a heavy-tailed one, namely Eq. (43).

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen, 2007) is conventionally optimized by stress majorization rather than gradient methods, which frequently fail to converge to good mappings; nonetheless, a few works employ gradient methods (Kruskal, 1964; Zheng et al., 2018), and we treat this case briefly. In particular, Zheng et al. (2018) formulates an effective gradient-descent MDS algorithm for graph drawing. We begin from the cost function, absorbing the weight $w_{i,j}$ into the learning rate without loss of generality:

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

The loss comprises a single term per pair, with no explicit separation into attractive and repulsive parts; each such term produces both attraction and repulsion. Because the loss is separable across the (i, j) components, we may analyze a single pair. Its derivative is

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

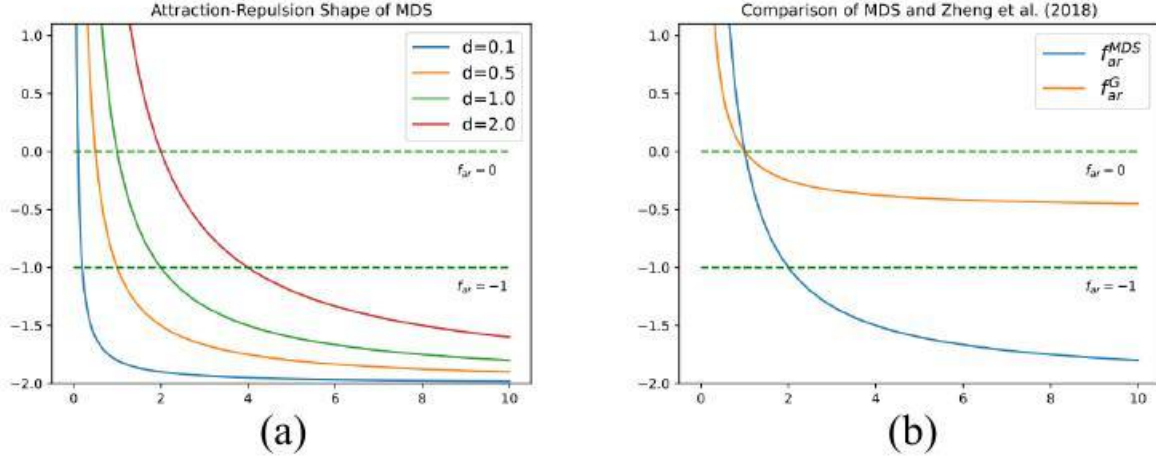


Figure 17: (a) Attraction–repulsion characteristic of MDS, given by f_{ar} . (b) Comparison of the attraction–repulsion characteristics of MDS and the method of (Zheng et al., 2018) at $d = 1.0$.

The attraction–repulsion shape of MDS is therefore

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. The shape is attractive for $\zeta > d$, where it is bounded to $[-2, 0]$ (Fig. 17(a)). This range is unsuitable for convergence, however, since only values in $[-1, 0]$ contract while the remainder expand, and values below -2 additionally cause endpoint crossover with expansion. To circumvent this, (Zheng et al., 2018) adopts an ad hoc formulation inspired by force-directed graph layout,

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

which is valid for an effective learning rate $\lambda \leq 1$: since f_{ar}^G is bounded to $[-0.5, 0]$, any $\lambda \leq 1$ ensures contraction. For $\zeta < d$, in contrast, $f_{ar}^{MDS} > 0$ and the interaction is repulsive. Overall, this attraction–repulsion interaction is most effective when the initialization is already close to the target. Continued optimization of a single pair oscillates about the distance $\zeta = d$ at which $f_{ar}^G = 0$, which plays a role analogous to UMAP’s ζ_{-1} . No learning rate drives this distance to zero; indeed, $\zeta = d$ is precisely the MDS objective. Consequently, MDS clusters are typically more diffuse than those of UMAP or PaCMAP (see (Lambert et al., 2022), (de Bodt et al., 2025), and (Kury et al., 2025) for illustrations). We note that (Lambert et al., 2022) reformulates the MDS loss as a quartet stress, defined over four samples via a relative distance (each distance normalized by the sum of the six pairwise distances in the quartet), affording global regularization and lower computational cost than majorization.

H Construction of the High-Dimensional Graph

Stochastic Neighbor Embedding (SNE) (Hinton and Roweis, 2002) underlies modern dimensionality reduction algorithms. Given a dataset $X = \{x_i \in R^n | i = 1, \dots, N\}$, it constructs the high-dimensional graph via the system

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_t - x_v\|_2^2 / 2\sigma_t^2)}, \quad (87)$$

where σ_i^2 is set so as to match a user-specified *perplexity* P , defined by

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

UMAP, by contrast, constructs its high-dimensional graph from the k -nearest-neighbor (k -NN) graph via the system

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ denotes the set of k nearest neighbors of x_i and σ_i is a scaling parameter chosen so that $\sum_j p_{i|j} = \log_2(k)$. The graph is then symmetrized by the probabilistic t-conorm

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP uses just the k -NN graph for the affinities (all equal to 1) with a self-tuning distance measure (Zelnik-Manor and Perona 2004),

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the mean Euclidean distance from x_i to its fourth through sixth nearest neighbors. This σ_i serves the same purpose as the corresponding scaling parameters in Eqs. (87) and (90), although all three are defined differently.

Irrespective of these graph-construction choices, our analysis identifies the optimization, and specifically the attraction and repulsion shapes, as the primary determinants of the low-dimensional embedding.

I Detailed Results for Varying λ_a and λ_b

This section reports detailed results for the experiments summarized in Fig. I(h,i). These results were obtained by substituting the attraction and repulsion shapes of UMAP with those of the other methods. Figure I(18) reproduces the main-text results. Figures I(19-22) display, for each method, the individual embeddings across the (λ_a, λ_r) grid, together with the initialization, a reference embedding under annealed λ , and the limiting cases $\lambda_a = 0$ and $\lambda_r = 0$. For PaCMAP, which additionally uses mid-near pairs, we include a reference embedding with the mid-near pairs enabled. Figures I(23-27) and Figs. I(28-32) report the corresponding results on the FMNIST and single-cell transcriptomes datasets, respectively.

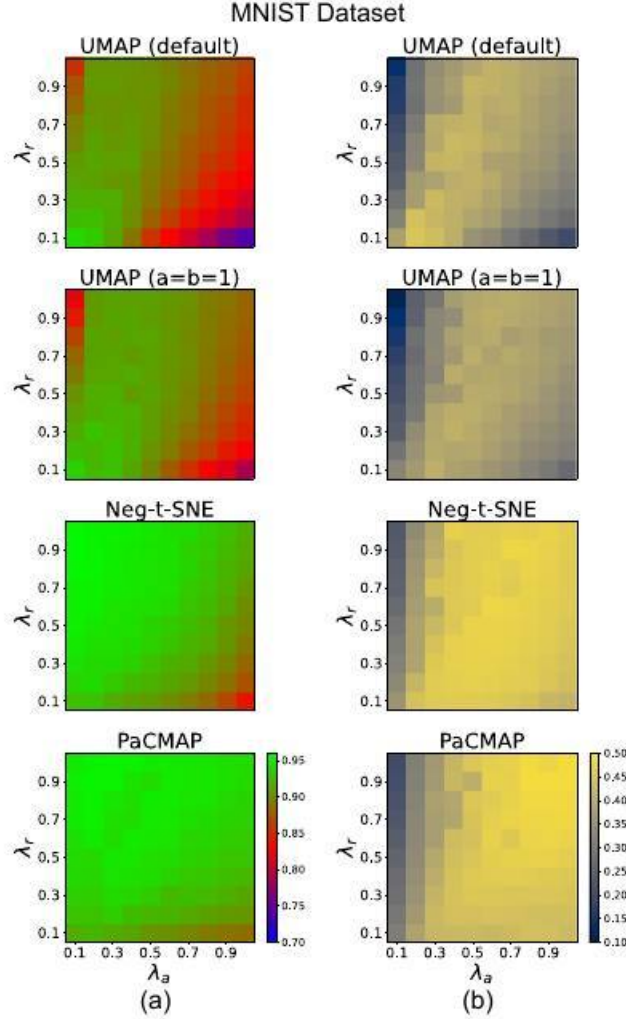


Figure 18: (a) Trustworthiness and (b) Silhouette score across methods for the MNIST dataset.

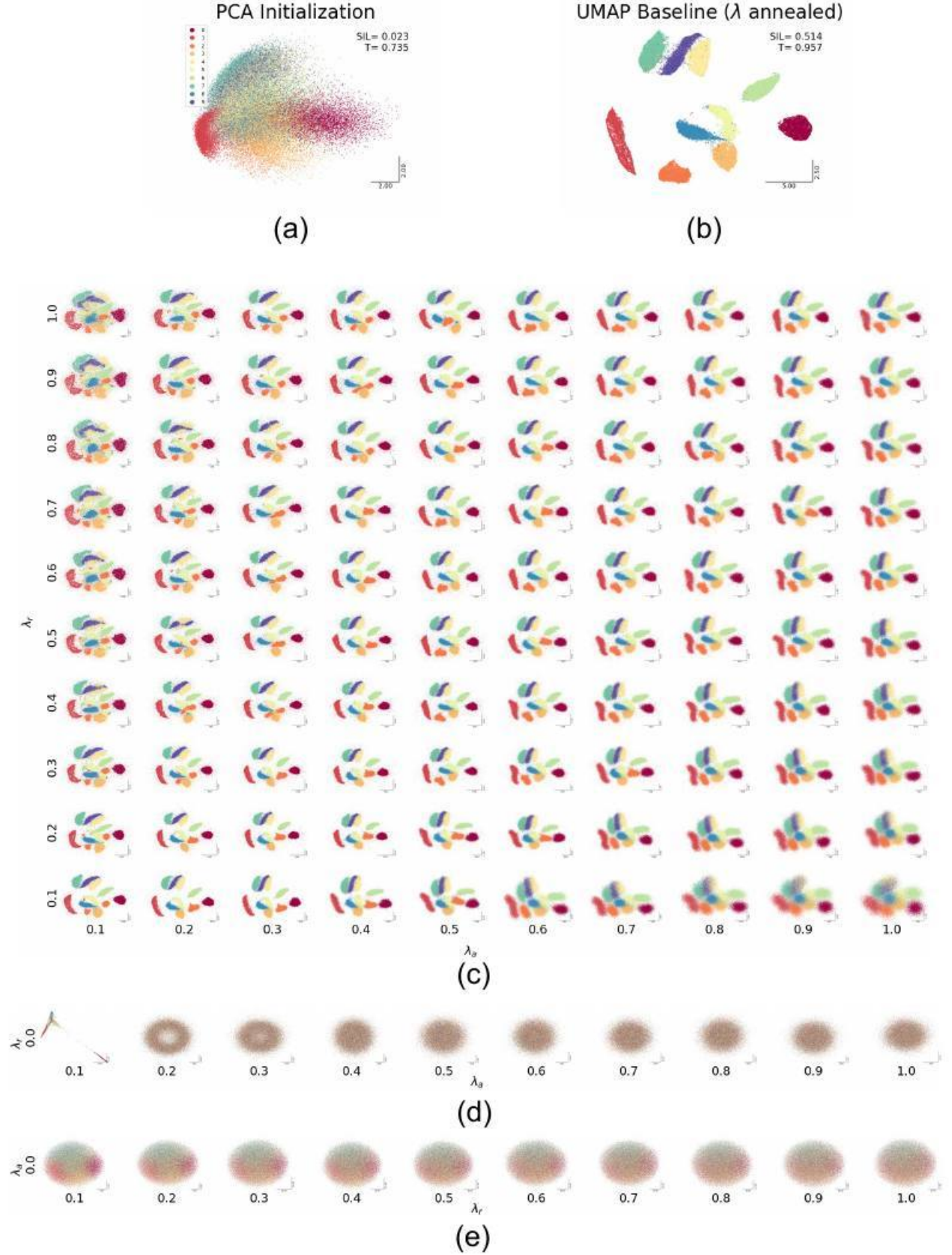


Figure 19: Varying λ_a and λ_r for the MNIST dataset using UMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Baseline obtained with λ annealed from 1. (c) Embeddings as λ_a and λ_r are varied, with no annealing. (d) At $\lambda_r = 0$ (no repulsion), the attractive force alone produces no clusters. (e) Symmetrically, at $\lambda_a = 0$ (no attraction), the repulsive force alone produces no clusters.³⁰

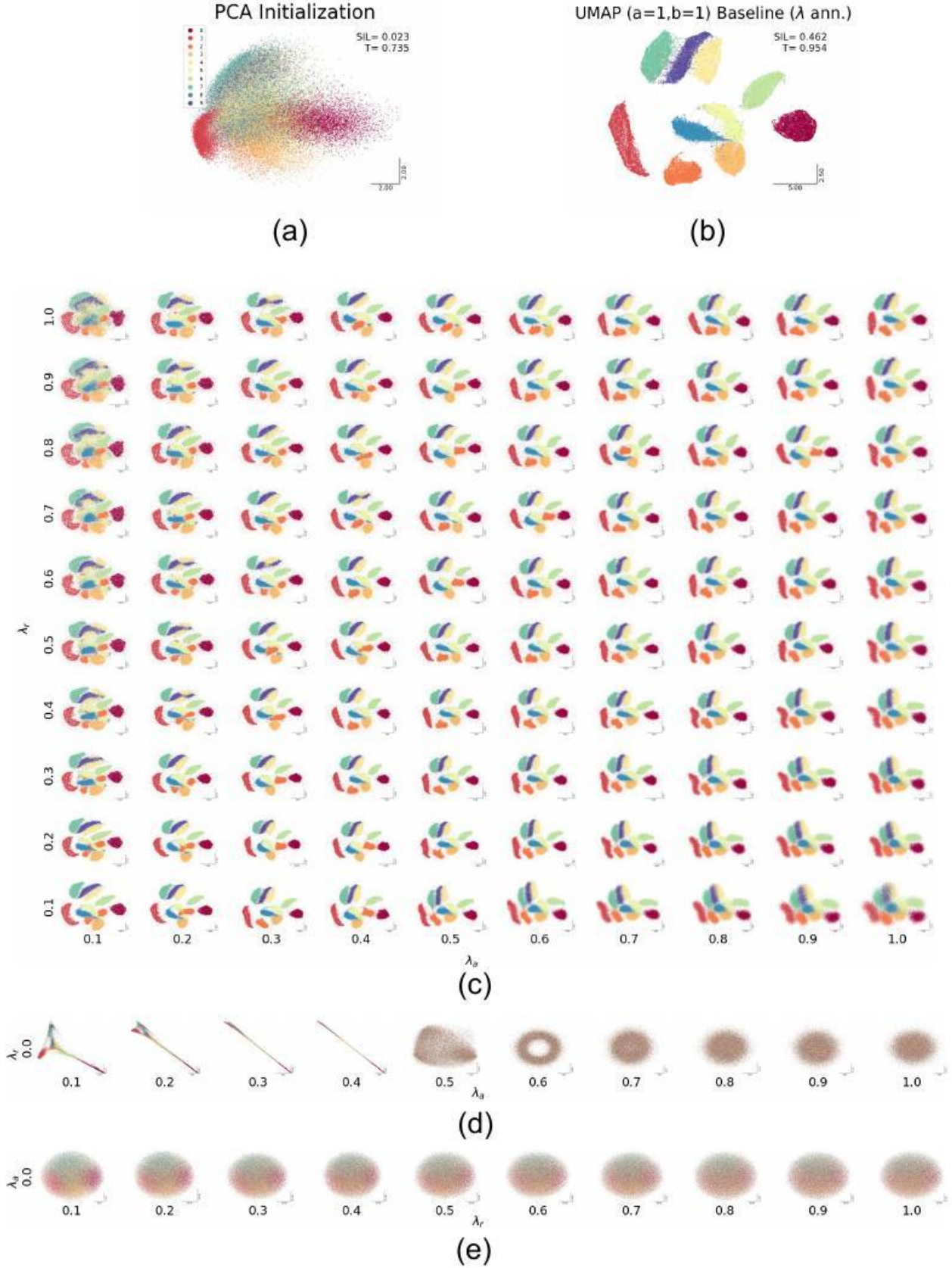


Figure 20: Varying λ_a and λ_r for the MNIST dataset using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Embedding initialization. (b) Baseline obtained with λ annealed from 1. (c) Embeddings as λ_a and λ_r are varied, with no annealing. (d) At $\lambda_r = 0$ (no repulsion), the attractive force alone produces no clusters. (e) Symmetrically, at $\lambda_a = 0$ (no attraction), the repulsive force alone produces no clusters.

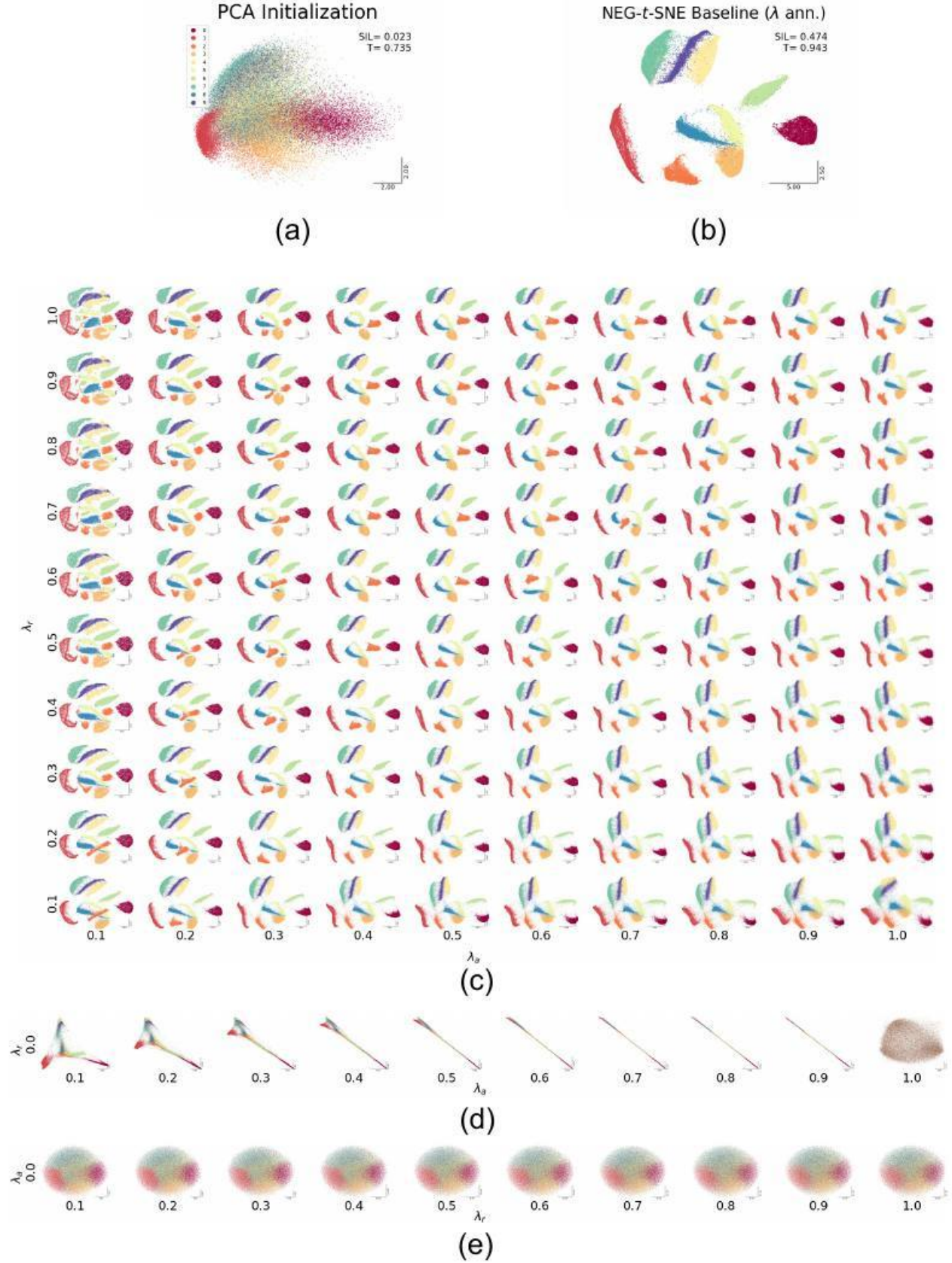


Figure 21: Varying λ_a and λ_r for the MNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Embedding initialization. (b) Baseline obtained with λ annealed from 1. (c) Embeddings as λ_a and λ_r are varied, with no annealing. (d) At $\lambda_r = 0$ (no repulsion), the attractive force alone produces no clusters. (e) Symmetrically, at $\lambda_a = 0$ (no attraction), the repulsive force alone produces no clusters.³²

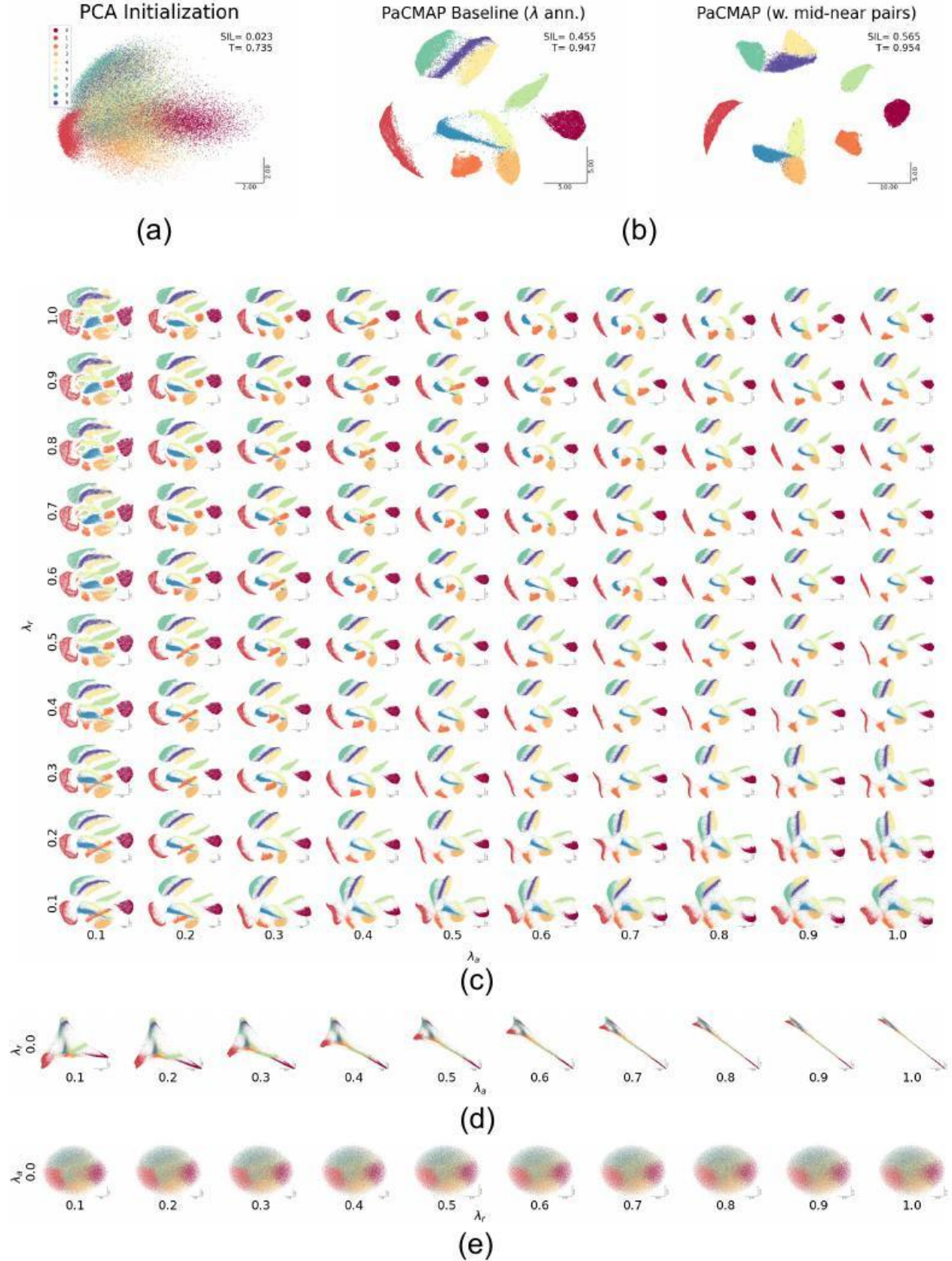


Figure 22: Varying λ_α and λ_r for the MNIST dataset using PaCMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Left: baseline with λ annealed from 1 (mid-near pairs excluded, to isolate the attraction–repulsion interaction); right: with mid-near pairs included. (c) Embeddings as λ_α and λ_r are varied, with no annealing. (d) At $\lambda_r = 0$ (no repulsion), the attractive force alone produces ≈ 10 clusters. (e) Symmetrically, at $\lambda_\alpha = 0$ (no attraction), the repulsive force alone produces no clusters.

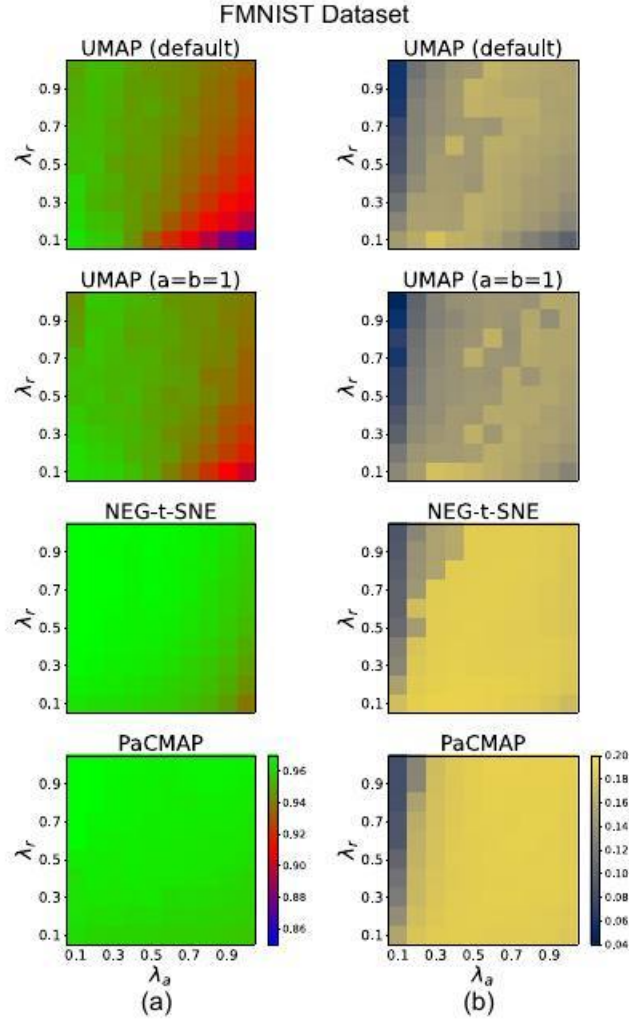


Figure 23: (a) Trustworthiness and (b) Silhouette score across methods for the FMNIST dataset.

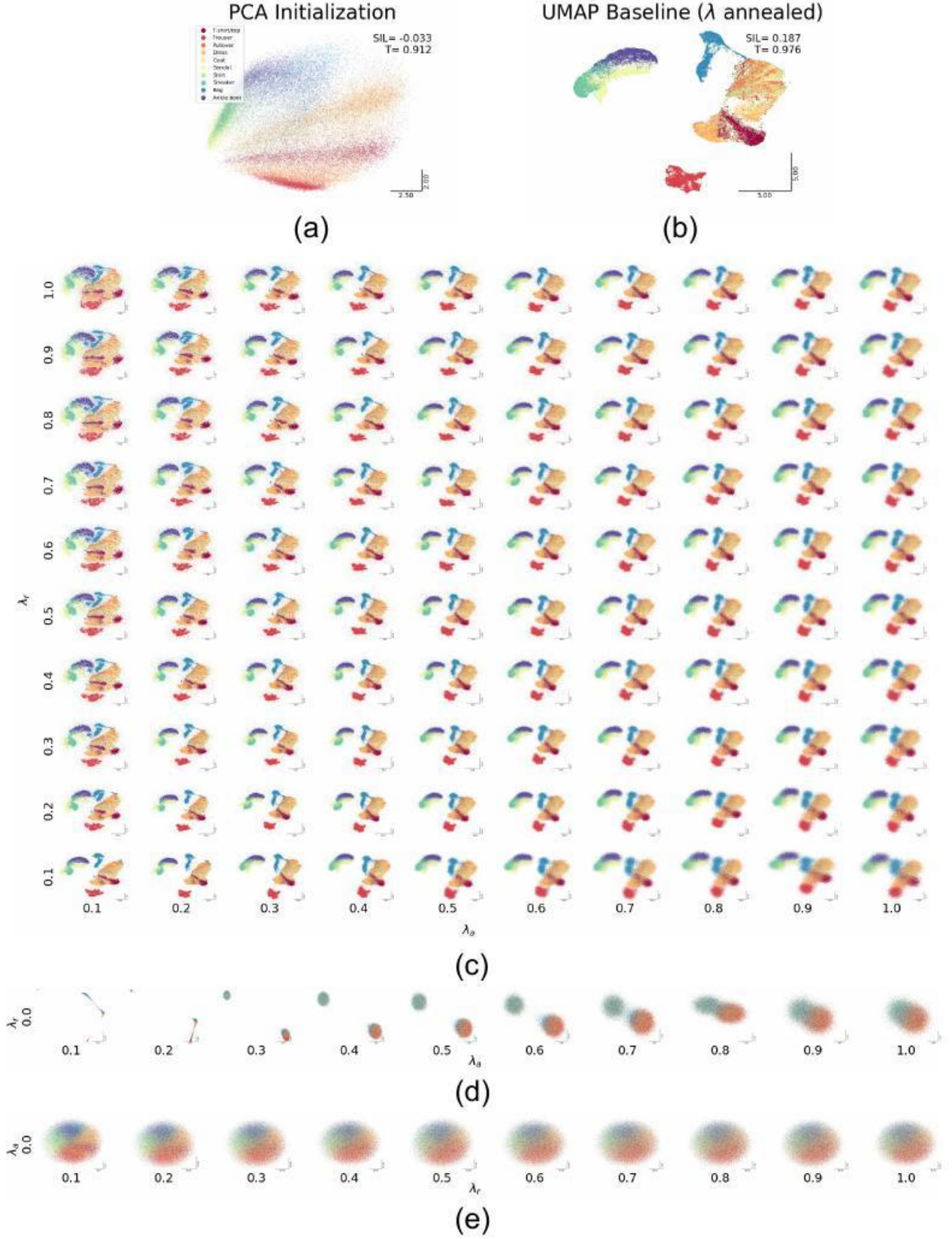


Figure 24: Varying λ_a and λ_r for the FMNIST dataset using UMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Baseline obtained with λ annealed from 1. (c) Embeddings as λ_a and λ_r are varied, with no annealing. (d) At $\lambda_r = 0$ (no repulsion), the attractive force alone produces no clusters. (e) Symmetrically, at $\lambda_a = 0$ (no attraction), the repulsive force alone produces no clusters.

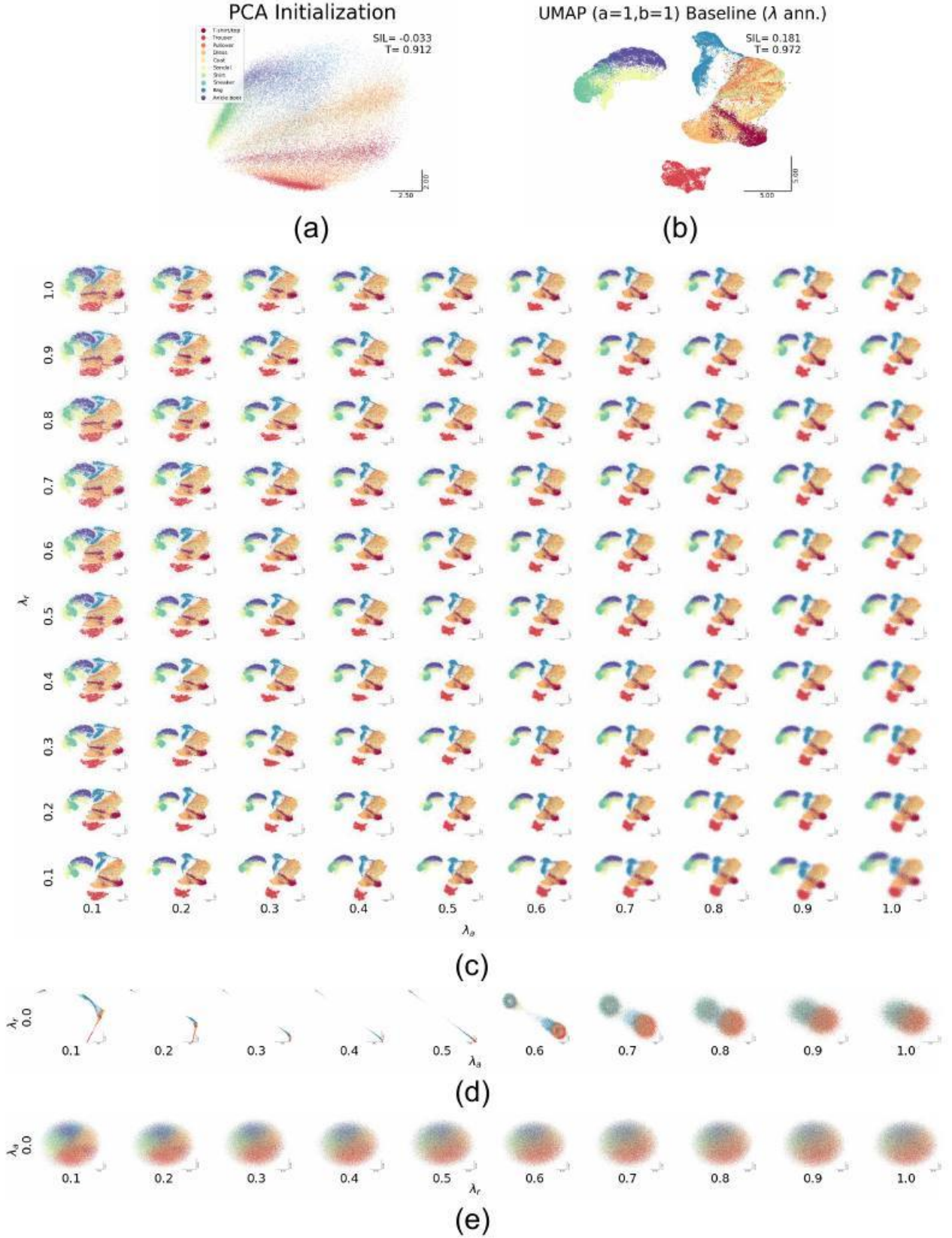


Figure 25: Varying λ_a and λ_r for the FMNIST dataset using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Embedding initialization. (b) Baseline obtained with λ annealed from 1. (c) Embeddings as λ_a and λ_r are varied, with no annealing. (d) At $\lambda_r = 0$ (no repulsion), the attractive force alone produces no clusters. (e) Symmetrically, at $\lambda_a = 0$ (no attraction), the repulsive force alone produces no clusters.

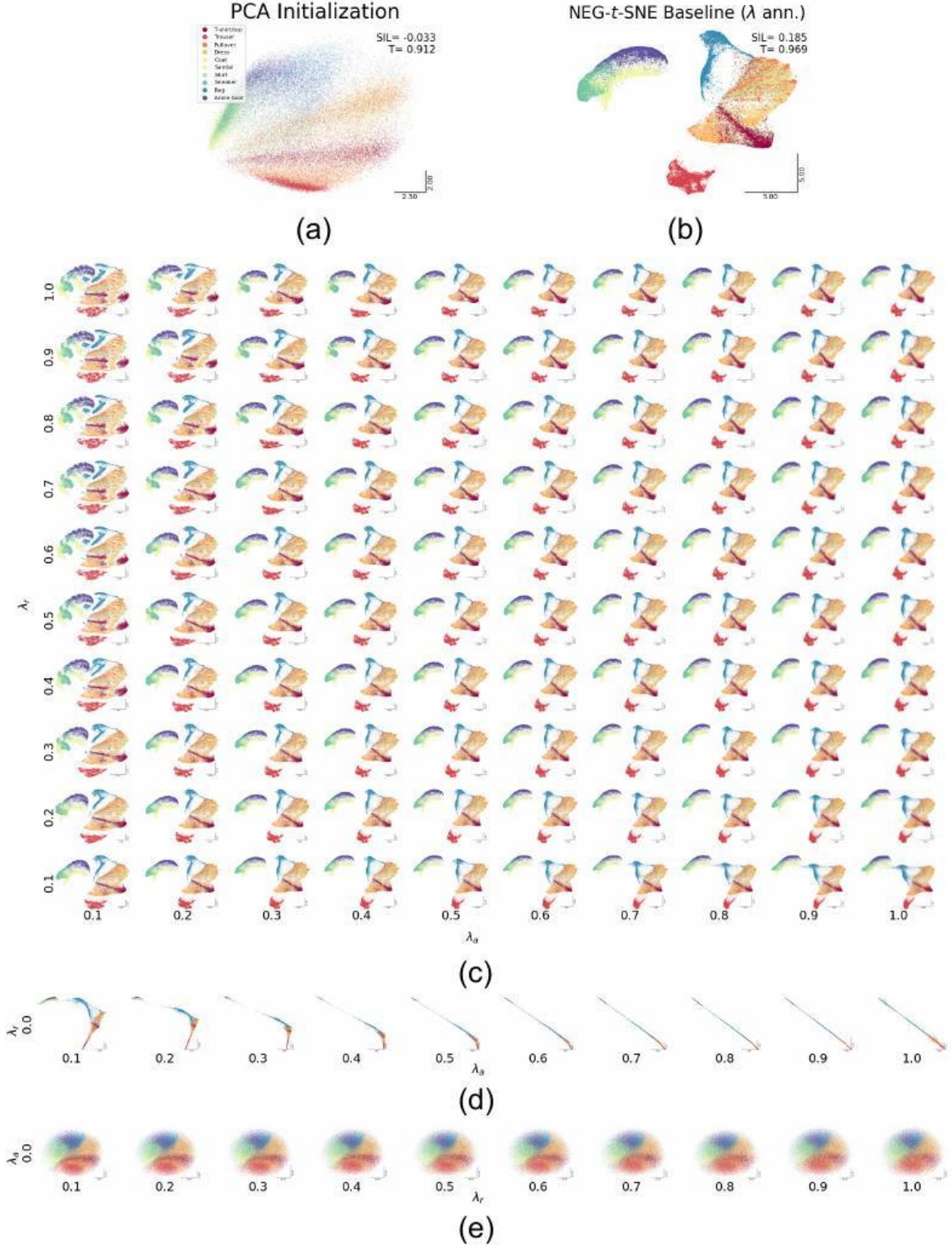


Figure 26: Varying λ_a and λ_r for the FMNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Embedding initialization. (b) Baseline obtained with λ annealed from 1. (c) Embeddings as λ_a and λ_r are varied, with no annealing. (d) At $\lambda_r = 0$ (no repulsion), the attractive force alone produces no clusters. (e) Symmetrically, at $\lambda_a = 0$ (no attraction), the repulsive force alone produces no clusters.

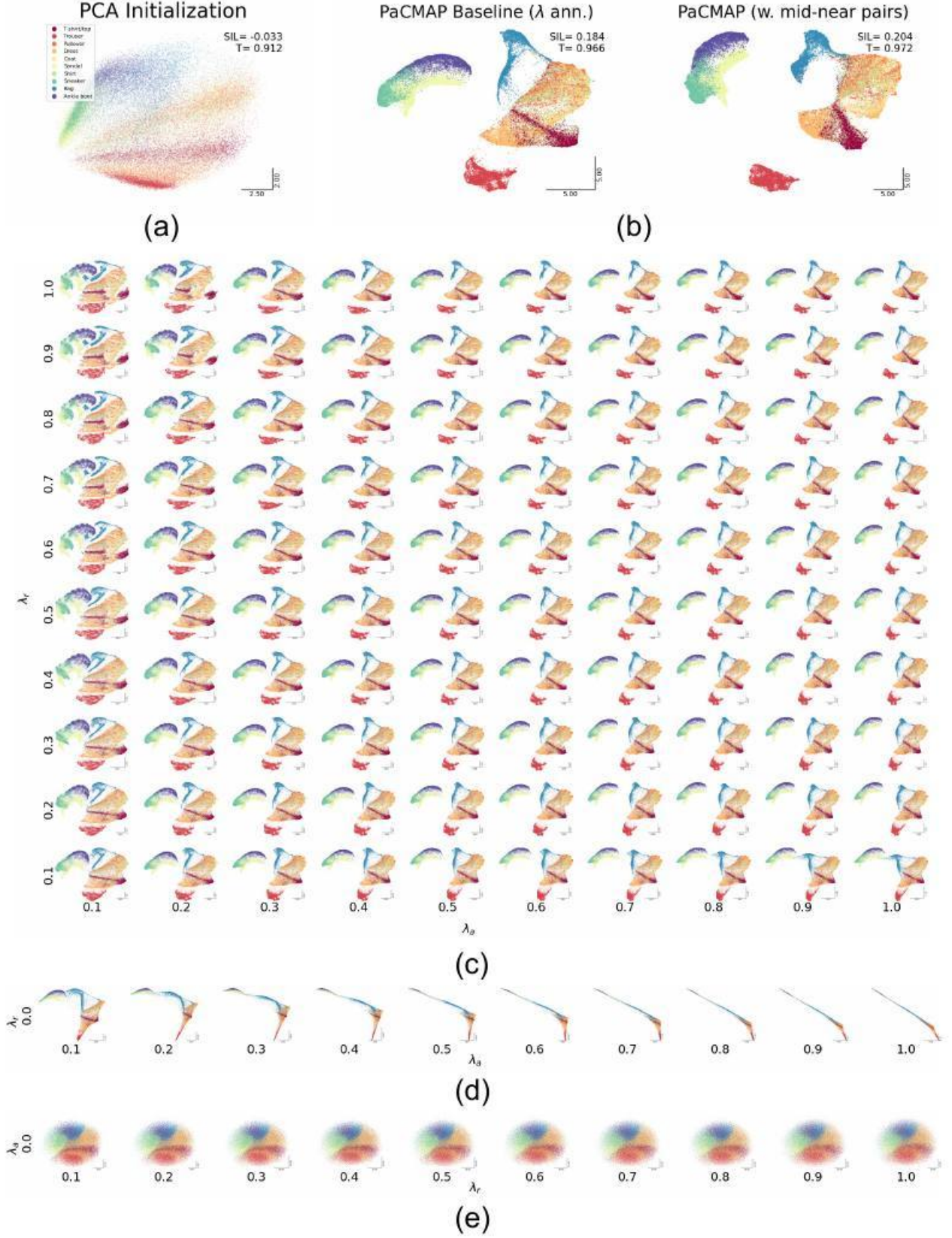


Figure 27: Varying λ_a and λ_r for the FMNIST dataset using PaCMAP’s attraction and repulsion shapes. (a) Embedding initialization. (b) Left: baseline with λ annealed from 1 (mid-near pairs excluded, to isolate the attraction–repulsion interaction); right: with mid-near pairs included. (c) Embeddings as λ_a and λ_r are varied, with no annealing. (d) At $\lambda_r = 0$ (no repulsion), the attractive force alone produces no clusters. (e) Symmetrically, at $\lambda_a = 0$ (no attraction), the repulsive force alone produces no clusters.

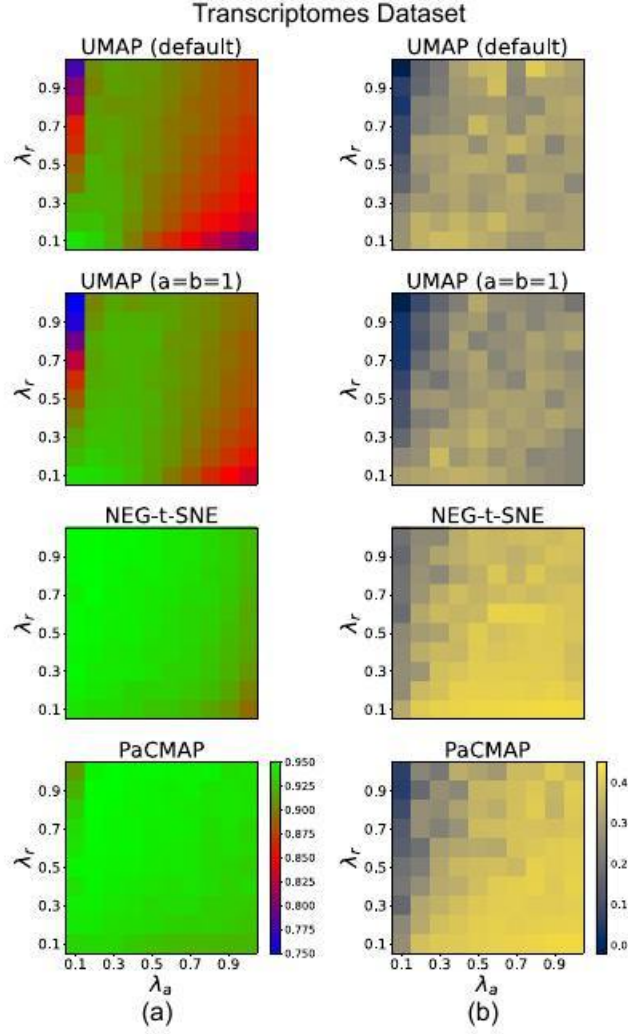


Figure 28: (a) Trustworthiness and (b) Silhouette score across methods for the single-cell transcriptomes dataset.

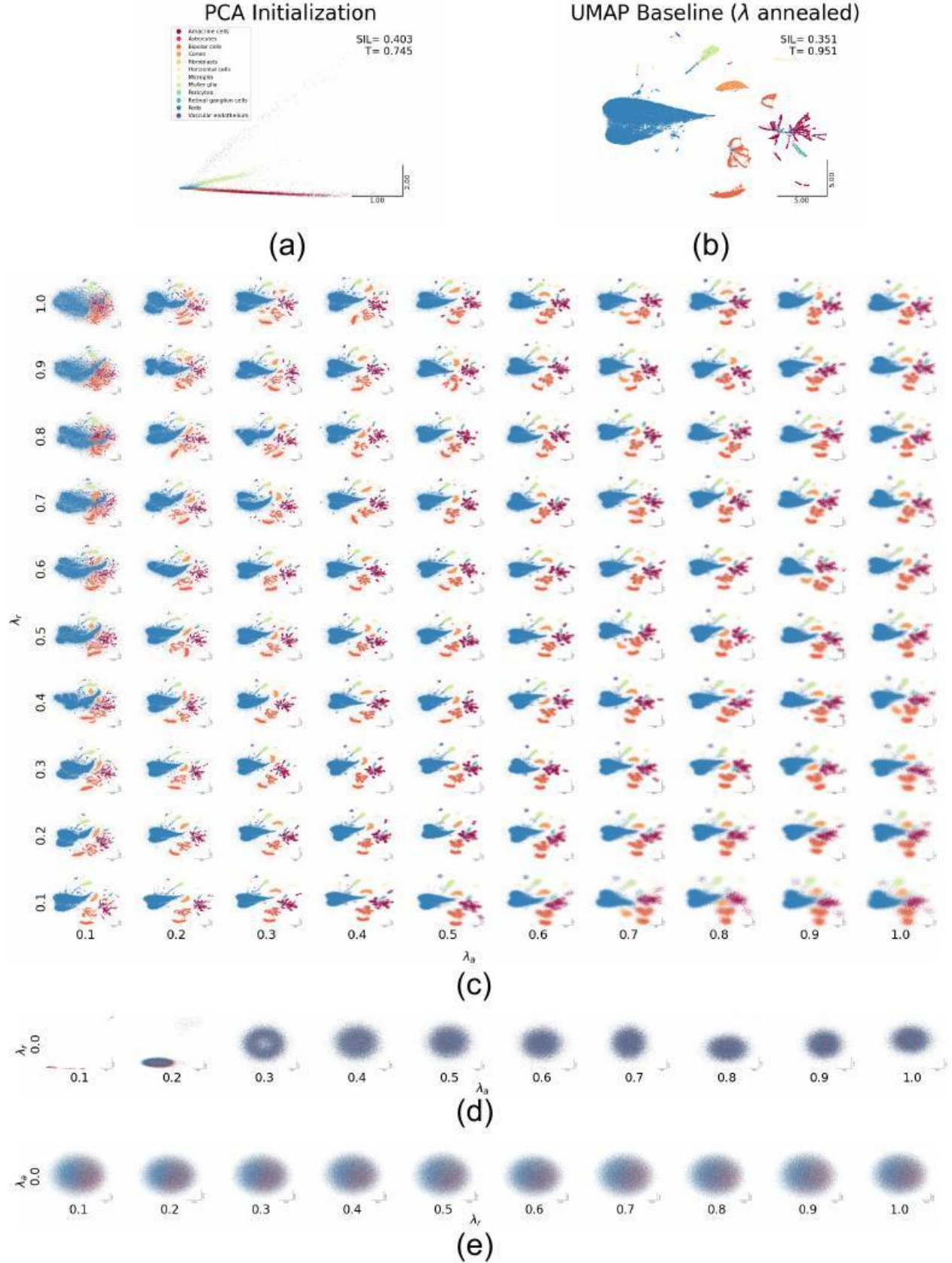


Figure 29: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Baseline obtained with λ annealed from 1. (c) Embeddings as λ_a and λ_r are varied, with no annealing. (d) At $\lambda_r = 0$ (no repulsion), the attractive force alone produces no clusters. (e) Symmetrically, at $\lambda_a = 0$ (no attraction), the repulsive force alone produces no clusters.

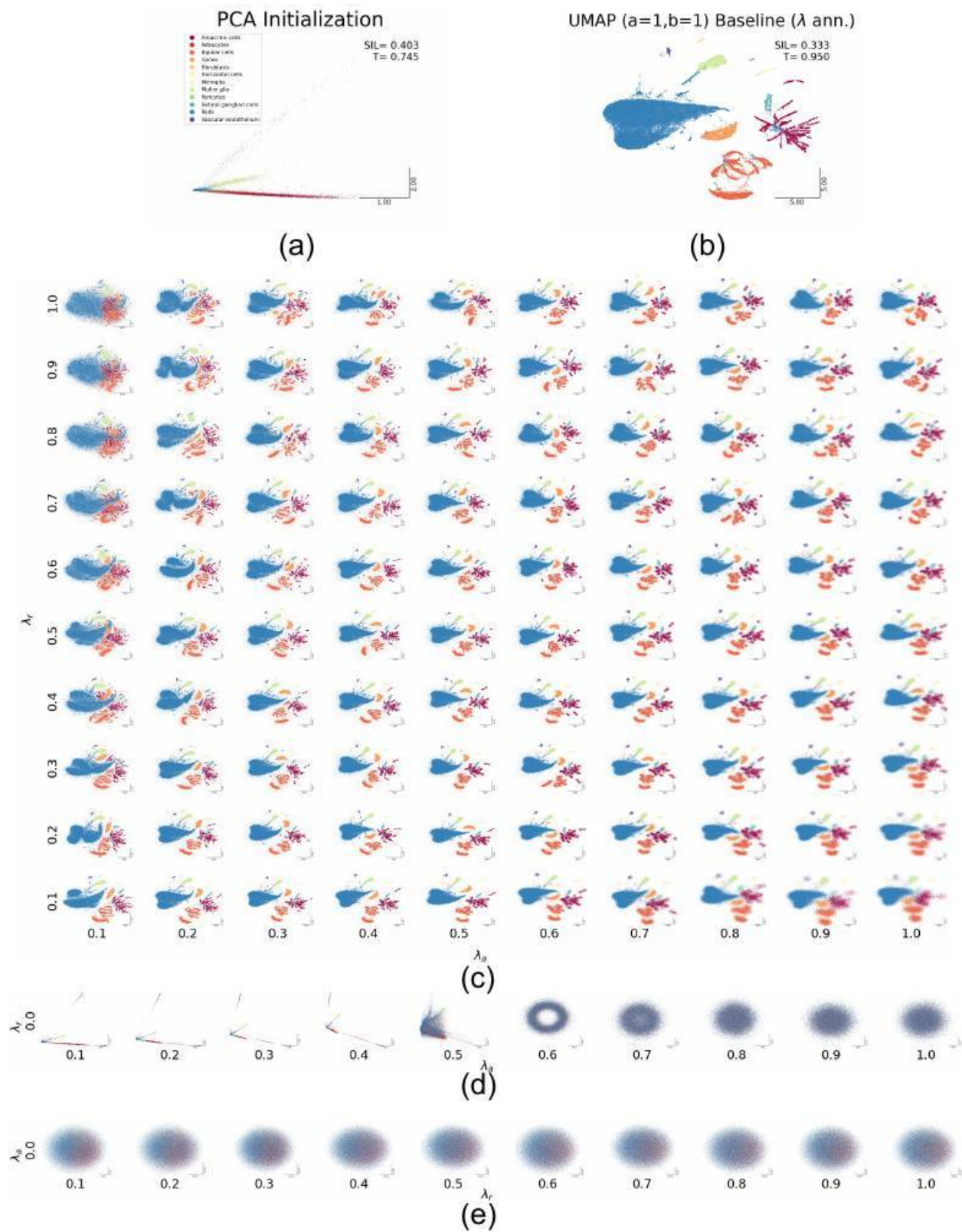


Figure 30: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Embedding initialization. (b) Baseline obtained with λ annealed from 1. (c) Embeddings as λ_a and λ_r are varied, with no annealing. (d) At $\lambda_r = 0$ (no repulsion), the attractive force alone produces no clusters. (e) Symmetrically, at $\lambda_a = 0$ (no attraction), the repulsive force alone produces no clusters.

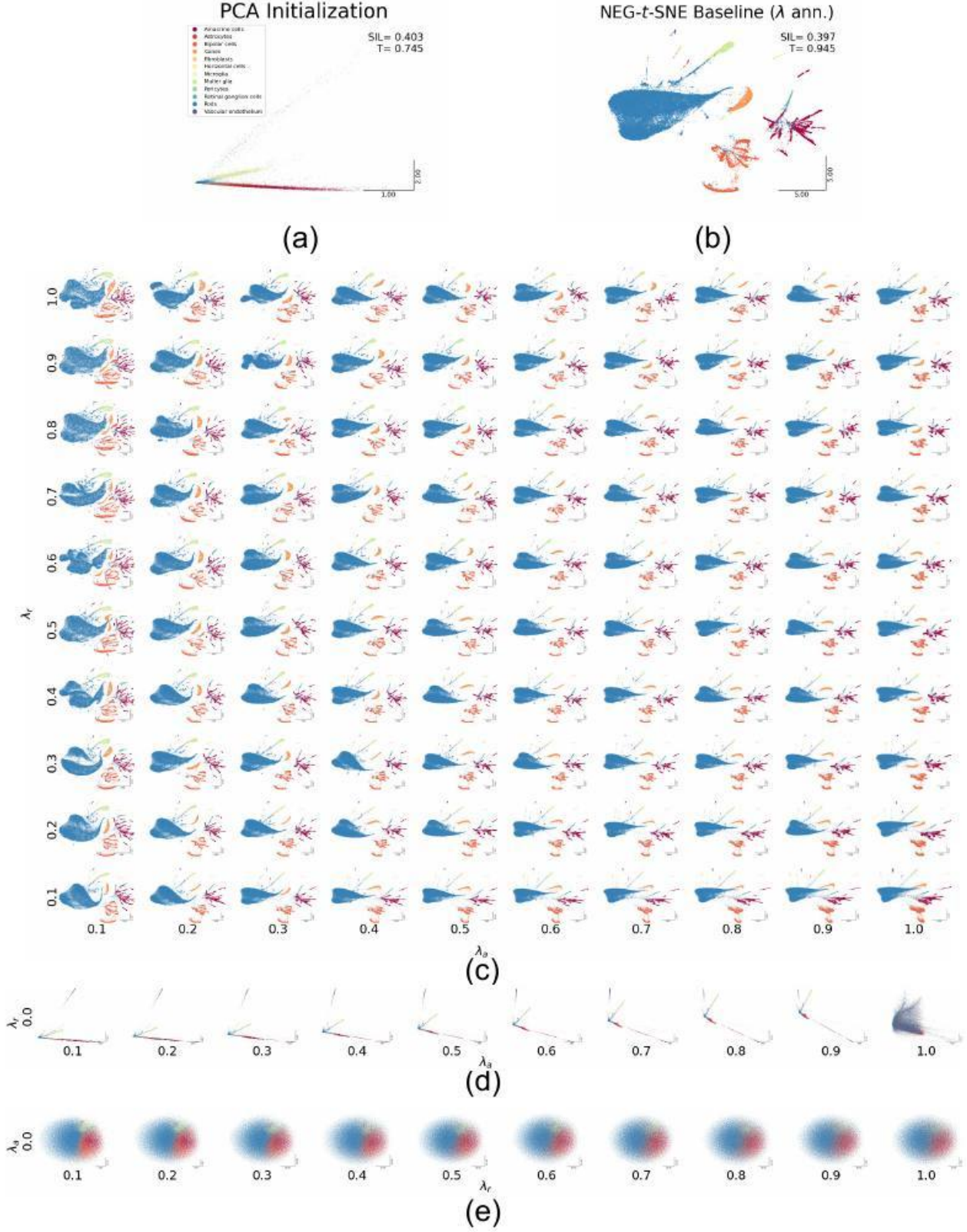


Figure 31: Varying λ_a and λ_r for the single-cell transcriptomes dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Embedding initialization. (b) Baseline obtained with λ annealed from 1. (c) Embeddings as λ_a and λ_r are varied, with no annealing. (d) At $\lambda_r = 0$ (no repulsion), the attractive force alone produces no distinct clusters. (e) Symmetrically, at $\lambda_a = 0$ (no attraction), the repulsive force alone produces no clusters.

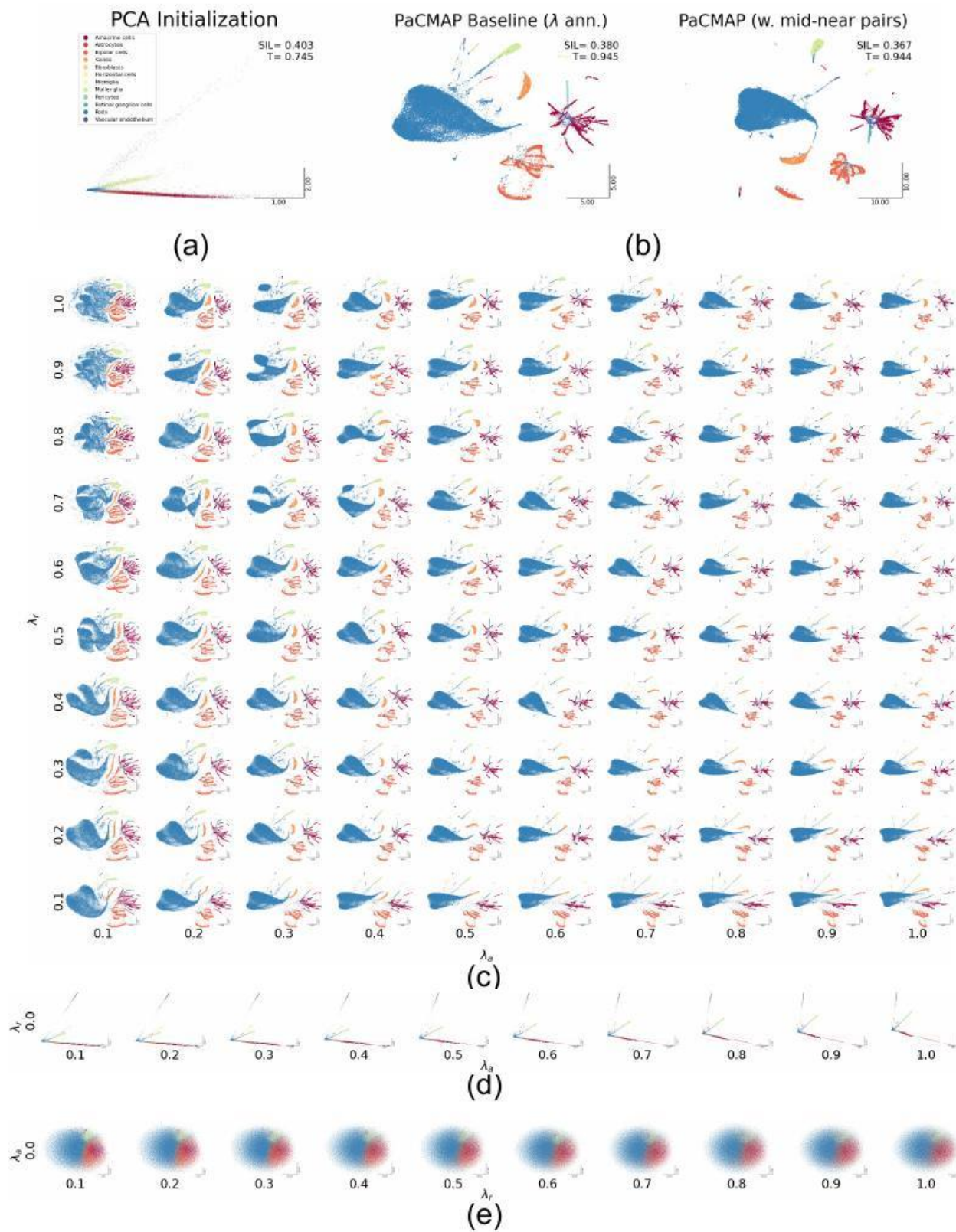


Figure 32: Varying λ_a and λ_r for the single-cell transcriptomes dataset using PaCMAP’s attraction and repulsion shapes. (a) Embedding initialization. (b) Left: baseline with λ annealed from 1 (mid-near pairs excluded, to isolate the attraction–repulsion interaction); right: with mid-near pairs included. (c) Embeddings as λ_a and λ_r are varied, with no annealing. (d) At $\lambda_r = 0$ (no repulsion), the attractive force alone produces no clusters. (e) Symmetrically, at $\lambda_a = 0$ (no attraction), the repulsive force alone produces no clusters.

J Mix and Match Approach

Our analysis treats the attraction and repulsion shapes as independent, which implies that shapes derived from distinct loss functions and design principles may be combined arbitrarily. Figure 33 presents several such combinations. In each case we restrict the attraction shape to $[-1, 0]$ (satisfying Proposition 4.1) to reduce its sensitivity to the learning rate, and we use only finite repulsion shapes to suppress large repulsive forces at small distances. To ensure that any observed difference is attributable to the shapes rather than to the learning-rate schedule or optimizer, we fix $\lambda = 1$ and use UMAP’s optimization procedure throughout. UMAP’s attraction shape (via $f_a = 0.5f_a^U$) yields superior clustering (higher silhouette score), whereas PaCMAP’s attraction shape yields superior local structure (higher trustworthiness); the latter follows from PaCMAP’s attraction shape lying in $[-0.5, 0]$, which precludes endpoint crossover during the updates. Confining the UMAP and NEG- t -SNE attraction shapes to $[-0.5, 0]$ (via $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$, respectively) improves the local structure at the cost of clustering performance (Fig. 34).

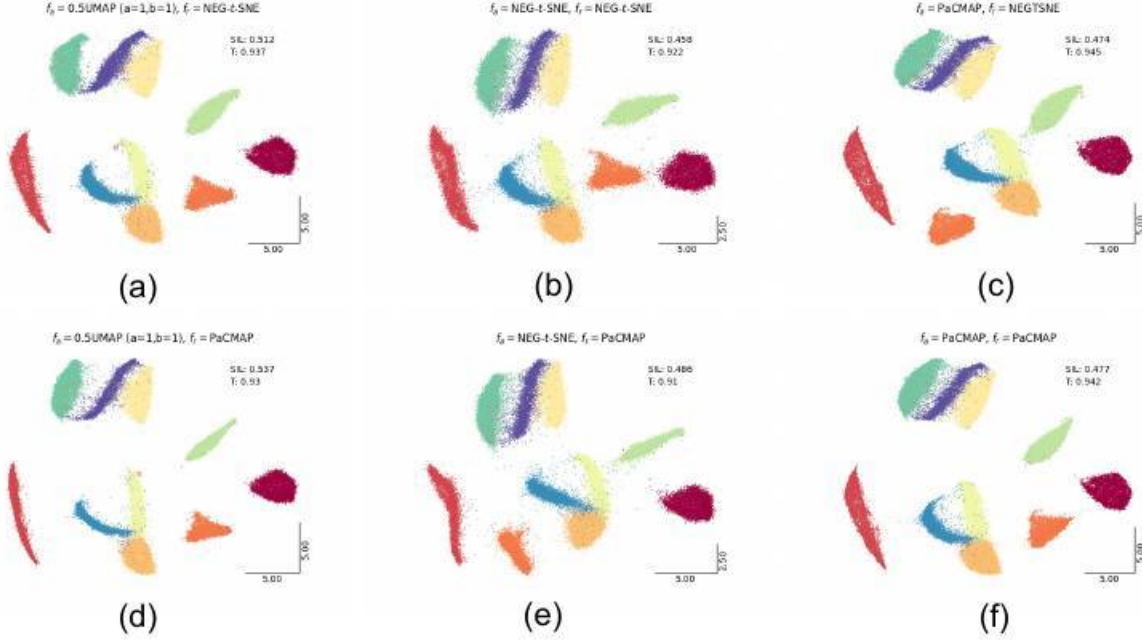


Figure 33: Embeddings of MNIST obtained by combining attraction and repulsion shapes across methods at a constant learning rate ($\lambda = 1$). The attraction shapes satisfy Proposition 4.1 and the repulsion shapes are finite. Top row: the UMAP, NEG- t -SNE, and PaCMAP attraction shapes, each paired with the NEG- t -SNE repulsion shape. Bottom row: the same attraction shapes, each paired with the PaCMAP repulsion shape.

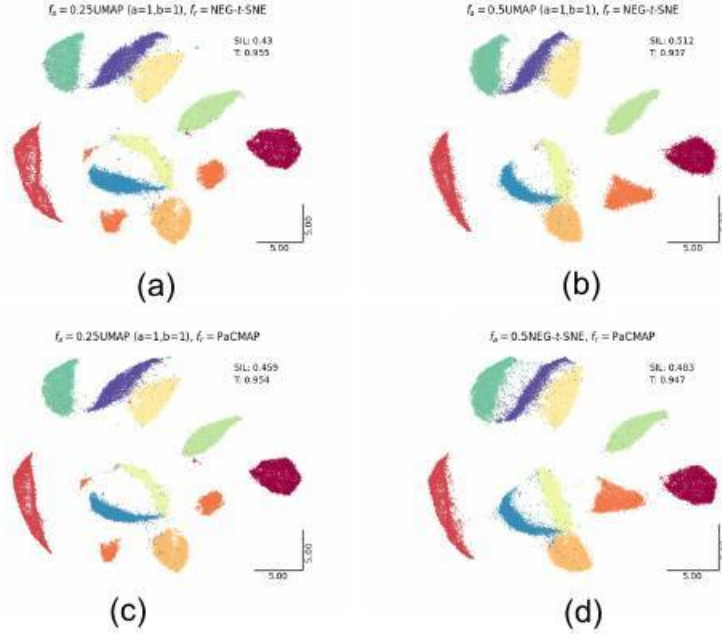


Figure 34: Embeddings of MNIST obtained by combining attraction and repulsion shapes across methods at a constant learning rate ($\lambda = 1$). The attraction shape is confined to $[-0.5, 0]$ to preclude endpoint crossover during the attractive update. Top row: the UMAP and NEG- t -SNE attraction shapes, each paired with the NEG- t -SNE repulsion shape. Bottom row: the same attraction shapes, each paired with the PaCMAP repulsion shape.

K Implementation Details

For the analysis we implemented UMAP from scratch. We used `numba` (Lam et al., 2015) to compute an exact (rather than approximate) nearest-neighbor graph with $k = 15$, and the `scikit-learn` (Pedregosa et al., 2011) PCA implementation for the PCA initialization. This implementation produced and quantified the embeddings in Figs. 1, 2, 3, 4, 6, 8, 9, and 10, and was reused with the attraction and repulsion shapes replaced by those of the alternative methods (Figs. 18, 34 unless stated otherwise). Trustworthiness and silhouette scores were computed with the corresponding `scikit-learn` functions.

The mappings in Figs. 2, 3, 9, 10, and 15 are rotated onto the respective reference embedding (panel (a) of each figure). The alignment is performed by normalizing each embedding to zero mean and unit norm and then extracting the rotation and scaling via SciPy’s (Virtanen et al., 2020) `orthogonal_procrustes` routine.

For the comparison with Neg- t -SNE in Fig. 11, both UMAP and Neg- t -SNE were run using the original implementation of the contrastive-embedding framework; the implementation details will be released upon acceptance.

The PaCMAP and LocalMAP embeddings in Figs. 15, 16, 22 (b) (right), 27 (b) (right), and 32 (b) (right) were obtained with the official PaCMAP package; the implementation details will be released upon acceptance.

The code for rendering the PubMed embeddings in Fig. 7 will be released upon acceptance.